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*Design and Implementation of a Multimodal
Dataset Structure and Data Acquisition Framework
for Dog Behavior Research*

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Author's Declaration

During the preparation of this work, the author used

- GPT-5.3 in order to: Grammar and spelling check, Improve writing style, Formatting assistance.
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Signed: *Giovanni Gallerani*

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“TOMORROW IS IN YOUR HANDS”

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Riassunto

Questa tesi è il risultato del tirocinio interno svolto presso il TUAT Biosignal Informatics Laboratory della Tokyo University of Agriculture and Technology, nel contesto del programma “Bando Atlante” organizzato dall’Università degli Studi di Ferrara.

Il cane è stato il primo animale a essere addomesticato dall’essere umano, un legame antico che precede la nascita dell’agricoltura. Oggi i cani continuano a ricoprire ruoli chiave nella nostra società, non solo come animali da compagnia, ma anche in ambito medico, terapeutico e nelle forze dell’ordine. Comprenderne lo stato emotivo è quindi diventato sempre più importante.

Negli ultimi anni, con l’emergere di tecniche di machine learning e data science, l’approccio tecnologico ha iniziato a svolgere un ruolo sempre più rilevante nello studio del comportamento animale.

Lo scopo di questa tesi è documentare il lavoro svolto e le conoscenze acquisite nella progettazione della struttura di un dataset multimodale per lo studio del comportamento canino, insieme allo sviluppo di un insieme di strumenti software dedicati all’acquisizione dei dati e dei metadati associati, utilizzati per popolarlo. Questo lavoro pone le basi per future ricerche scientifiche mirate a ridurre la distanza tra noi e il mondo animale che ci circonda.

Introduction

This thesis work was carried out at TUAT (Tokyo University of Agriculture and Technology) [23] within the framework of the “Bando Atlante” program [2], an opportunity offered to students of the Università degli Studi di Ferrara (Unife) to study and undertake traineeships at extra-European universities.

During the traineeship and thesis research period, I contributed to the project “Creation of an AI Interaction Platform for the Coexistence of Humans, Animals, and AI”, developed at the TUAT Biosignal Informatics Laboratory [25]. The project is part of the CRONOS (Cutting-edge Research and Development on Information & Communication Science) program [41], a research initiative managed by the Japan Science and Technology Agency (JST) [40].

The long-term objective of this project is the development of a communication service enabling the AI-based estimation of internal states of animals based on non-invasive measurements, such as video recordings or ECG signals. The ultimate goal is to enable humans, animals, and artificial intelligence to interact within a unified framework, forming a “Human and Animal in the Loop” (HAITL) interaction platform capable of understanding animal behavior, psychology, and physiology, and responding appropriately. Such a system aims to reduce the difficulties humans experience when interpreting the internal states of animals [17].

As part of the effort to develop this platform, the laboratory is currently focusing on human-dog interaction analyzing several publicly available datasets containing non-invasive measurements collected from dogs. In parallel, significant effort is being devoted to the creation of a new in-house multimodal dataset for dog behavior research. This dataset will serve as the foundation for future research projects involving machine learning models aimed at inferring dogs' internal states from non-invasive measurements, as well as to provide additional data to current researches already performed inside the laboratory. In addition, the dataset is intended to be released publicly in the future in order to facilitate further research on dog behavior by the broader scientific community.

The present thesis documents the practical application of the knowledge and skills acquired during the research stay at TUAT. In particular, the work focuses on two main contributions developed during the traineeship period: (i) the design of the dataset structure for the new multimodal dataset currently under development at the laboratory, and (ii) the design and development of a data acquisition framework used to populate the dataset with experimental data and their associated metadata.

The dataset structure was designed following the Brain Imaging Data Structure (BIDS) standard [38], while the data acquisition framework was implemented using a set of Python programs based on the Lab Streaming Layer (LSL) framework [34], which enables synchronous acquisition of data from multiple sensors connected to different computers during recording sessions. The framework is complemented by additional Python tools for supporting the management of the dataset contents and inserting additional metadata.

The remainder of this thesis is structured as follows. First, the problem of understanding the internal states of dogs is introduced, together with an overview of recent progress in the field ([chapter 2](#)). Next, the experimental design is described, including the characteristics of the collected data, the experimental setup, and the data acquisition procedure ([chapter 3](#)). This is followed by a formal introduction to the BIDS standard and a detailed description of the proposed multimodal dataset structure ([chapter 4](#)). Subsequently, the design and implementation of the software tools developed for dataset creation is presented together with a usage example ([chapter 5](#)). Finally, the thesis concludes with a discussion of the contributions of this work and possible future developments ([chapter 6](#)).

Given the early-stage nature of this project, some aspects of the data acquisition process may evolve over time in response to the laboratory's research needs. Therefore, the description provided in this thesis reflects the experimental procedures and data collection practices adopted by the TUAT Biosignal Informatics Laboratory as of March 2026.

In accordance with the principles of Open Science, all software developed within the context of this thesis has been made publicly available on GitHub [13].

Preliminaries

2.1 A New Frontier in Dog Behavioral Studies

The relationship between humans and dogs is ancient. Dogs were the first species to be domesticated by humans, with estimates placing the initial domestication event between 15,000 and 20,000 years ago, even before the development of agriculture [31]. Today, dogs continue to play a fundamental role in human society, not only as companion animals but also in fields such as healthcare, therapy, and law enforcement. Understanding the emotional and mental states of both species has therefore become increasingly important. While much attention has been devoted to understanding whether dogs are capable of interpreting human emotions, it is equally important to consider the inverse question.

Dogs appear to be sophisticated readers of human emotions. Research has shown that dogs are capable of cross-modal emotional processing, integrating visual cues (such as facial expressions), auditory cues (such as vocal tone), and even olfactory information (in the form of chemosignals) when interpreting human emotional states. Additionally, dogs often use their owners as a reference point to guide their behavior in ambiguous situations, and they tend to look more frequently at their owners than at unfamiliar humans when seeking information [28, 36].

On the contrary humans appear to have more difficulty correctly interpreting the emotional signals expressed by dogs. Studies show that people often misinterpret canine emotional cues, frequently confusing fearful facial expressions with happiness and bodily expressions of fear with sadness. Moreover, this misinterpretation often occurs without awareness. In one study, 96% of participants believed they could correctly identify happiness in dogs, while their actual accuracy was only 63%. The discrepancy is even more pronounced for fear: although 90% of participants believed they could recognize it, the correct identification rate was only 16% [30]. These results suggest that the issue lies not in the lack of communicative signals from dogs, but rather in a mismatch between the signals produced by dogs and the way humans interpret them.

The consequences of the COVID-19 pandemic have also highlighted how the human–dog relationship can influence behavior on both sides. During this period, humans and dogs spent significantly more time together due to lockdown restrictions. This increased proximity led to several positive outcomes, such as more frequent playtime, increased time spent together, and more regular dog walking, all habitude that remained also after the end of the pandemic. These changes were associated with a reduction in certain behavioral problems in dogs, including aggression toward family members. However, a slight increase in separation anxiety and fear-related behaviors was also observed [43].

These findings demonstrate how changes in human lifestyle can significantly influence canine behavior. If dogs are capable of adapting their behavior in response to changes in human routines, it becomes increasingly important for humans to develop better tools for understanding canine emotional states and behavioral signals.

In recent years, technological approaches have started to play an increasingly important role in this area of research. In particular, machine learning and data science techniques are being applied to improve the automatic analysis and interpretation of dog behavior.

Several studies have explored different types of data sources. For example, deep learning techniques have been applied to 3D pose estimation for recognizing stress-related behaviors [45]. Other approaches rely on 3D cameras for movement tracking [42], accelerometer-based motion data for behavior classification [27], or physiological signals such as heart rate variability [33].

With the growing success of multimodal fusion models in human behavior recognition [37], recent studies have begun to combine multiple modalities in the analysis of dog behavior as well. For instance, some works integrate video and motion data to improve behavioral recognition performance [32].

Following this emerging research direction, the TUAT Biosignal Informatics Laboratory has initiated efforts to develop multimodal datasets for dog behavior analysis. Given the current lack of datasets combining video, audio, motion, and biosignal data, the laboratory has started acquiring the necessary sensors and designing an experimental protocol for data collection (chapter 3) with the goal of developing an in house dataset to be used as basis for some of the laboratory researches.

This reason motivated the development of a structured dataset format (chapter 4) together with a dedicated data acquisition framework for organizing and recording the collected data (chapter 5). The design and implementation of these systems constitute the main contribution of this thesis.

Experimental Design

This chapter describes the experimental design associated with the creation of the multimodal dog behavior dataset. It begins with an overview of the types of data collected in the dataset ([section 3.1](#)), followed by a description of the laboratory setup used for the experiments ([section 3.2](#)). Next, the participant recruitment procedure is presented ([section 3.3](#)). Finally, the experimental protocol used for data acquisition is described together with the ethical considerations adopted to ensure the welfare of the dogs participating in the study ([section 3.4](#)).

3.1 Collected Data Modalities

From the early conceptualization stage of the project, the collection of a large and diverse set of data describing human–dog interaction was identified as a key requirement of the study. The resulting dataset is therefore multimodal and integrates several types of data recorded simultaneously during each experiment. These modalities are rarely available together in publicly available datasets on dog behavior.

At the time of writing, the dataset includes four main types of data acquired directly during each recording session:

- **Video data:** video recordings capturing the dog’s behavior obtained through the usage of multiple cameras positioned around the experimental environment, allowing the scene to be observed from different angles.
- **Audio data:** audio recordings obtained using wireless microphones. One microphone is attached to the dog’s neck in order to capture barking and other vocalizations produced by the animal during the experiment. Two additional microphones are used to record the dialogue between the researcher and the owner. During the experiment, the researcher periodically asks the owner to evaluate the internal state of the dog using valence and arousal scores based on the Self-Assessment Manikin (SAM) scale [26]. These scores range from 1 to 9 and together represent the estimated emotional state of the dog (Figure 3.1). Valence represents the affective state of the dog along a spectrum ranging from sadness to happiness, whereas arousal represents the level of physiological activation, ranging from low activation to high excitement or panic.

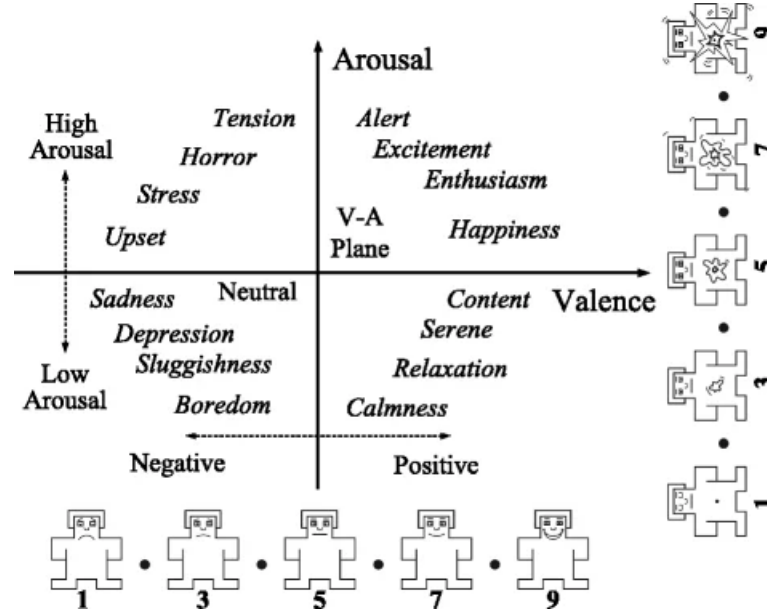


Figure 3.1: Representation of emotions in the valence–arousal space.

- **Motion data:** time-series data recorded using four wearable inertial measurement unit (IMU) worn by the dog, containing measurements from a

gyroscope and an accelerometer.

- **Physiological (biosignal) data:** time-series data representing the electrical signal measured from the dog’s electrocardiogram (ECG) two devices, each connected to a pair of electrodes.

To acquire these data, an experimental protocol was designed in which both the dog and its owner interact within a series of predefined scenarios. Each scenario corresponds to a specific task intended to stimulate particular interaction patterns or behavioral responses. The details of the data acquisition procedure are described in [section 3.4](#).

In addition to the data directly recorded during the experiments, further information can be derived through dedicated data processing pipelines. The main examples of this are:

- **Time-series data describing the dog’s internal state:** valence and arousal scores associated with specific time intervals of the recording can be derived from the owner assessments recorded during the experiment. It is possible to obtain also experts annotations by having veterinarians analyze the video recordings and periodically evaluate the valence and arousal levels associated with different segments of the experiment.
- **Motion data derived from video:** three-dimensional keypoints describing the dog’s body movement can be estimated from the video recordings by combining information obtained from multiple camera views.

3.2 Experimental Setup

3.2.1 Laboratory Environment

For each dog participating in the experiments, behavioral and physiological data were recorded inside a dedicated laboratory space located within the TUAT Koganei Animal Medical Emergency Center [14]. The layout of the laboratory environment is illustrated in the following figure ([Figure 3.2](#)).

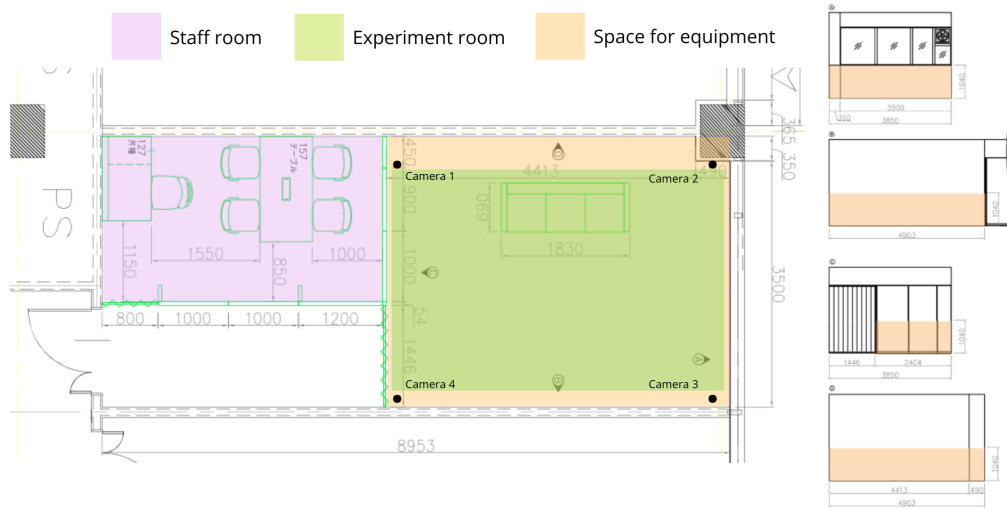


Figure 3.2: Planimetry of the laboratory room used for the experiments.

As shown in the planimetry, the laboratory space can be divided into three main areas:

- **Staff room:** this area is used by the researchers during the experiment. It contains the computers used for data acquisition as well as cleaning equipment used to restore the experimental room to its original condition after each recording session.
- **Experiment room:** this is the main area where the experiments are conducted and where data are recorded using multiple sensing devices. The floor of the experimental area whose dimensions are approximately 360 x 300 cm is divided into 30 squares measuring 60×60 cm each. This grid is used to assist in estimating the three-dimensional position of the dog's body keypoints from the recorded video data. The calibration grid view from each camera is shown in the following figure (Figure 3.3).

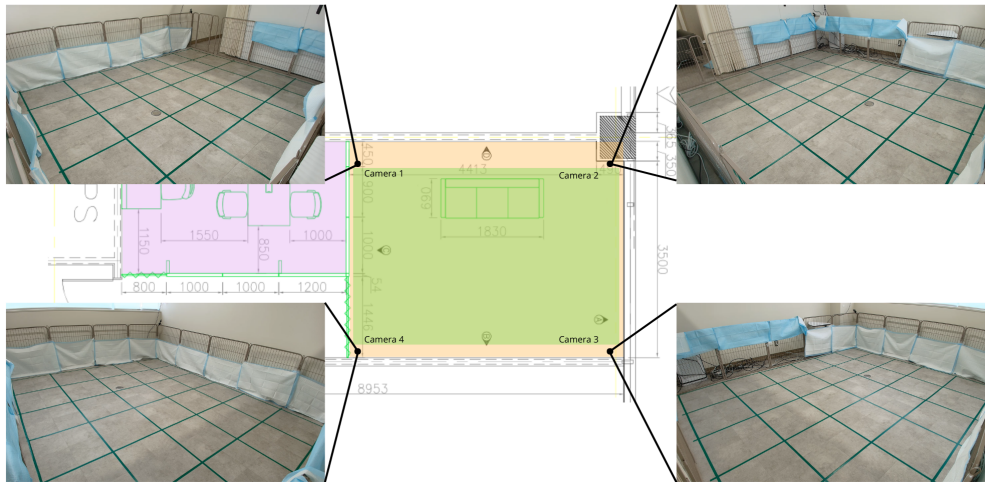


Figure 3.3: Calibration grid view from each camera in the experiment room. (photo taken before positioning the sofa)

The planimetry also shows the location of the four cameras used to record video data of the dog's behavior. The cameras are positioned at the four corners of the room and oriented toward the center of the experimental area, with a downward tilt that ensures the entire body of the dog remains within the camera frame.

A sofa is placed along the western wall of the room and is used by the dog owner during the experiment. The experimental area is enclosed by a metallic fence that separates it from the surrounding equipment area. Access to the experiment room is provided through a lockable gate integrated into the fence. A curtain is placed near the entrance in order to isolate the experimental environment during recording sessions.

The northern side of the room includes a ventilation shaft that allows the air in the environment to be regularly refreshed and prevents the accumulation of odors.

On the eastern wall, opposite the sofa, a printed representation of the Self-Assessment Manikin (SAM) scale [26] is attached (Figure 3.4). This visual aid is provided to help the owner to evaluate the valence and arousal scores for the dog when prompted by the researcher during the experiment.

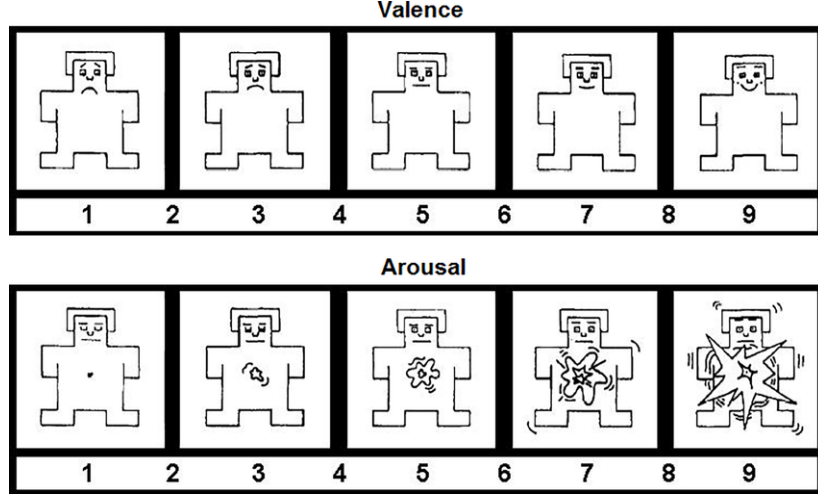


Figure 3.4: Self-Assessment Manikin scale used by the owner to evaluate valence and arousal during the experiment.

- **Space for equipment area:** a narrow area surrounding the experimental room used to house the cables connecting the cameras to the computers located in the staff room. This area is separated from the experiment room by the metallic fence to prevent dogs from accidentally disconnecting the equipment during the experiment.

3.2.2 Recording Devices

All experimental data are acquired using multiple sensors, including the previously mentioned cameras, connected to two G-Tune P6-I7G60BK-A portable computers [12], each equipped with 64 GB of RAM and used for data recording. The two computers are connected to the same local network and communicate using the Lab Streaming Layer (LSL) framework [34] to synchronize the data acquisition process.

Video data are recorded using four Logitech MX Brio cameras [6]. The cameras are configured to record at a frame rate of 30 frames per second with a Full HD resolution of 1920×1080 pixels. The camera outputs are distributed evenly across the two computers (two cameras per device) in order to avoid performance bottlenecks during the recording of high-resolution video streams.

Audio data are recorded using two Hollyland LARK MAX 2 wireless microphone systems [15]. Each system provides two wireless microphones whose audio signals are transmitted to a single receiver connected to a computer. One microphone is attached to the dog in order to capture barking and other vocalizations produced during the experiment. The two microphones of the second system are both used, one by the researcher and the other by the dog owner to record verbal interactions and the owner's assessments of the dog's internal state. All the microphones records with a sample rate of 44.1 kHz.

Motion and physiological data are recorded using a custom device developed by Motoi Noda, a second-year master's student and member of the TUAT Biosignal Informatics Laboratory. This device integrates two sensors models within a single unit.

The first sensor is an MPU6050 [19], an inertial measurement unit (IMU) that combines a three-axis gyroscope and a three-axis accelerometer, sampling motion data at 100 Hz. Four of this sensors are present in the device. The second sensor is an AD8232 [1], a single-lead heart-rate monitoring front-end designed to extract, amplify, and filter small biopotential signals such as ECG. The ECG signal is sampled at a frequency of 300 Hz. Two sensors of this kind are used inside the device.

The device used for measuring motion and ECG data is attached to the dog using a custom-made wearable bib designed to maintain stable contact between the sensors and the animal's body, as illustrated in the following figure (Figure 3.5). The device connects to one of the laboratory computers via Bluetooth and transmits the recorded data at a rate of 300 Hz, corresponding to the highest sampling frequency among the two sensors. Since the motion sensor operates at a lower sampling frequency (100 Hz), its values are repeated across three consecutive transmitted samples in order to align them with the higher transmission rate used for the ECG signal.



Figure 3.5: Dog wearing the custom bib used for motion and physiological data acquisition.

A summary of the sensors and recording settings discussed in this section is provided in the following table (Table 3.1). The integration of these sensors within the data acquisition framework will be described later in this work (chapter 5).

Modality	Device	Sampling	Resolution/Channels	#Used
Video	Logitech MX Brio	30 fps	1920 × 1080 (Full HD)	4
Audio	Hollyland LARK MAX 2	44.1 kHz	Single channel	2 (3 mic)
Motion	MPU6050 IMU	100 Hz	6 channels (3 gyro., 3 accel.)	4
ECG	AD8232	300 Hz	1 channels	2

Table 3.1: Overview of sensors used for data acquisition and their main technical characteristics.

3.3 Participant Recruitment

Participants were recruited on a voluntary basis, with dog owners invited to participate in the experiment together with their dogs. To raise awareness and attract volunteers, a set of promotional flyers was distributed (Figure 3.6).

In order for a dog to be accepted as a subject for the study, several requirements had to be satisfied. First, the dog had to be in good health and at least one year

old. In addition, the dog had to be large enough to comfortably wear the custom bib used for physiological data acquisition.

This size constraint was introduced because the ECG sensor integrated in the wearable device requires sufficient contact with the dog's chest area in order to acquire reliable electrocardiogram signals. Dogs weighing less than approximately 3 kg generally do not provide enough chest surface to ensure stable electrode placement. For this reason, only medium-to-large dogs were considered suitable for participation in the experiment.



Figure 3.6: Recruitment flyer used to attract volunteers for the experiments.

3.4 Experimental Protocol

In order to ensure both the quality of the collected data and the ethical treatment of the animals involved in the study, the experimental sessions are divided into two types: *presessions* and *scenario sequences*.

The first sessions involving a given dog are referred to as *presessions*. During these sessions the owner and the dog enter the experimental room and are allowed

to freely interact, without receiving specific instructions from the researchers. Additionally members from the research group can freely interact with the dog in this sessions. The purpose of these sessions is to allow the dog to gradually become familiar with the experimental environment, the laboratory personnel, and the wearable sensors attached to its body.

The pre-session phase is designed to avoid starting the actual experiment from a condition of stress or discomfort for the dog. Such conditions would not only negatively affect the well-being of the animal, but would also reduce the quality and reliability of the collected data during the subsequent experimental sessions since it would alter the animal behavior.

During pre-sessions, a variable amount of data may be recorded. Video data are always recorded, while audio, motion (gyroscope and accelerometer), and ECG signals are recorded only when the dog appears sufficiently comfortable wearing the sensors. A veterinarian is present during all sessions in order to monitor the animal's condition and assist in the habituation process. If the stress level of the dog exceeds the level considered safe by the veterinarian, the experiment is immediately suspended.

After the dog has attended several pre-sessions and has become accustomed to the environment and the equipment, the main experimental phase can begin. These sessions are referred to as *scenario sequences*. During scenario sequences all sensors are used simultaneously, laboratory personnel cannot enter in the experiment room, and the owner is asked to follow specific instructions provided by the researchers.

Each scenario sequence is structured as an alternation of active interaction tasks and resting-state periods. At the beginning of the sequence, the owner is asked to perform one of the available interaction tasks with the dog for a duration of five minutes. The task is randomly selected between two possible activities: engaging in conversation with the dog or giving the dog treats.

This active phase is followed by a resting-state period lasting five minutes, during which the owner remains seated on the sofa without interacting with the dog while the dog remains in a resting condition. After this resting phase, the remaining interaction task that was not previously performed is carried out for an additional five minutes. The sequence concludes with a second five-minute resting-state period.

During the active interaction tasks, the owner is periodically asked by the researchers to evaluate the internal state of the dog using valence and arousal scores. These evaluations are performed with the help of the Self-Assessment Manikin poster displayed on the wall of the experimental room.

At the end of each five-minute task or resting period, the recording is stopped and a new run is started. Longer experimental sessions can be obtained by repeating the same task–rest structure multiple times. When sequences longer than four scenarios are performed, short pauses may be introduced between repetitions of the protocol in order to ensure that the dog remains comfortable and relaxed throughout the experiment.

In the next chapter ([chapter 4](#)), the structure of the multimodal dataset used to organize the data collected during the experiments will be described in detail.

Dataset Structure and Standardization

This chapter serves two main objectives. First, it provides a formal introduction to the Brain Imaging Data Structure (BIDS) standard ([section 4.1](#)), outlining the foundational concepts that informed the design of the Dog Behavior Dataset. Second, it presents the final structure of the multimodal dataset developed in this thesis, including the methodological considerations and challenges encountered in adapting the dataset to the BIDS framework ([section 4.2](#)).

4.1 The BIDS Standard

4.1.1 Overview of BIDS

The Brain Imaging Data Structure (BIDS) is a community-driven standard constituted by a set of guidelines for

- organizing, naming, annotating, and describing data files collected from a growing range of neuroscience modalities,

- creating statistical modeling of neuroimaging data,
- developing applications created to be used with BIDS data.

It provides a formalized specification for directory organization, file naming, and metadata description based on structured folder hierarchies, and encoding of files and metadata in JSON and TSV formats with controlled vocabularies [29, 3].

Prior to the introduction of BIDS, there was no widely accepted consensus on how to organize and share data obtained in neuroimaging experiments. Even within the same research laboratory, researchers could opt to arrange their data in different ways, leading to misunderstandings and inefficient data reuse, due to the time consuming process of rearranging data or adapting scripts to different directories layouts, significantly hindering reproducibility and collaboration.

The sharing of experimental data is a worthy goal, but only if the data are shared in a way that makes them FAIR (Findable, Accessible, Interoperable, and Reusable) [44]. A clear and formal organization scheme for data reduces the burden of curation for researchers and data sharing repositories, minimizes the likelihood of errors caused by misinterpretation of data, and enables the development of automated analysis tools that can leverage the dataset structure to analyze data with minimal user intervention. Additionally, standardized organization allows datasets to be automatically validated against the specification, ensuring structural compliance and improving overall data quality.

The problems related with the disjointed ways to organize neuroimaging data, and the perceived advantages related with the creation of a standard, created the incentives for the development of BIDS.

The birth of BIDS can be traced back ultimately to a social media post by Russ Poldrack on October 17 2014 [20, 39] referring to a talk that Chris Gorgolewski had given at the weekly Stanford cognitive/neuroscience seminar (“Frisem”). A reply to the post by Stuart Buck, that at the time was a program officer at the Laura and John Arnold Foundation, led to a discussion about a new way of sharing data that ultimately resulted in a substantial grant from the Foundation to the Stanford group, with the aim of developing a new data sharing platform that would improve scientific collaboration efforts. This new platform would ultimately become the OpenNeuro archive [35], and the support from the Arnold Foundation would help start the work on BIDS [39].

Since its first publication in 2016 [29], BIDS has enabled the standardization of many processes in neuroscience research and fostered a data ecosystem that promotes reproducible research, and makes collaboration on analysis and data sharing easy across research groups and institutions through standardized formats that are platform-independent and openly documented. The success of this solution has been testified by its wide adoption. The standard has been adopted by a multitude of labs around the world and by major data repositories, it is currently used in 2000 open datasets across repositories, including OpenNeuro, and continues to expand every day [4].

A key characteristic of BIDS is its strong community-driven development model. The standard is developed by and for the neuroscience community, and continues to evolve in response to emerging research needs and feedback thanks to its many contributors. Extensions to the specification are introduced through BIDS Extension Proposals (BEPs), community-driven processes which follow an established review process and ensure backward compatibility. This extensibility allows BIDS to support an expanding range of modalities while maintaining consistency across versions. A full list of finalized BEPs and BEPs in progress can be found on the extensions section on the BIDS website [16].

The standard is further supported by growing an ecosystem of tools, including validators and analysis applications (“BIDS Apps”), which works seamlessly on BIDS-compliant datasets. Built-in validation tools enable automatic verification of dataset structure and metadata completeness, ensuring that developed datasets comply with BIDS standards, thus reinforcing data quality and reproducibility [24]. The validator is available for use within several different environments to best suit individual user preferences and use cases, including a web browser based version [5] that was used for the purpose of this thesis.

BIDS currently supports a broad range of neuroimaging modalities, including functional magnetic resonance imaging (fMRI), electroencephalography (EEG), magnetoencephalography (MEG), intracranial EEG (iEEG), positron emission tomography (PET), and behavioral data [3]. Its conceptual richness allows it to accommodate diverse experimental paradigms while maintaining structural consistency. Given the breadth of the BIDS specification, only the concepts and structural principles directly relevant to the design of the Dog Behavior Dataset will be presented in the following sections.

4.1.2 Core Principles and Hierarchical Organization

A BIDS dataset consists of a collection of folders and files that are organized and named in compliance with the BIDS specification [38]. Before describing the implementation of the BIDS standard in details, it is required to introduce some terms that are used throughout the specification and will be referred in the rest of this work.

Starting with the most high level concepts, the following definitions are associated with all experiments supported by BIDS:

- **Modality:** the category of brain data recorded by a file. For MRI data, different pulse sequences are considered distinct modalities, such as T1w, bold or dwi. For passive recording techniques, such as EEG, MEG or iEEG, the technique is sufficiently uniform to define the modalities eeg, meg and ieeg. Behavioural experiments with no neural recording are also included in the possible modalities.
- **Participant:** a person or animal participating in the study, it is used interchangeably with term Subject. Since in the BIDS standard the collection of subjects is saved in the file participants.tsv, to avoid any source of confusion, only the term participant will be used for the rest of this work.
- **Session:** a logical grouping of neuroimaging and behavioral data consistent across participants. Session is often synonymous to a visit in a longitudinal study, or to multiple visits where different data types are obtained (for example fMRI on one day followed by DWI the day after). Sessions are used to organize the data when several identical or similar data acquisitions are planned and performed on all or most subjects, especially in the case of some intervention or change happening between sessions (for example, the dog getting used to the experimental environment and researchers after each visit).
- **Task:** a set of structured activities performed by the participant during the data acquisition, where “data acquisition” is intended as a continuous and uninterrupted block of time during which data was acquired according to a particular scanning sequence/protocol. Tasks are usually accompanied

by stimuli and responses, and can greatly vary in complexity, from keeping a resting state (which it is considered a task by the specification) to be involved in complex activities. Only one task can be tied to a data acquisition, therefore even if during the acquisition the subject performed multiple conceptually different behaviors (with different sets of instructions) they will be considered as part of the a single task.

- **Run:** a repetition of an uninterrupted data acquisition using the same acquisition parameters and having the same associated task. Given its definition, run is a synonym for data acquisition as intended by BIDS. It is important to specify that “uninterrupted” may look different by modality due to the nature of the recording. For example, in MRI or MEG, if a subject leaves the scanner, the acquisition must be restarted introducing a new run, while for some types of PET acquisitions, a subject may leave and re-enter the scanner without interrupting the scan. In the context of the study described in this work ([chapter 3](#)), each run is equivalent to a continuous camera acquisition associated with a specific task.

By using these concepts, a **Dataset** in BIDS is defined as a set of neuroimaging and behavioral data acquired from one or more participants, possibly from multiple sessions, for the purpose of a particular study.

The concepts of participant, session, task and run, among with others that are specific to the chosen modality of a study, are used in the BIDS standard for naming files and dataset folders. These concepts can be associated with a file trough the usage of **Entities**, elements that are used as filename components in the form of a hyphen-separated (‘-’) key-value pairs, contributing to the file identification. Each entity is characterized by three attributes: name, key and value type. Their definitions are listed below.

- **Name:** a comprehensive name describing the context of information to be provided via the entity (e.g. “Participant”, “Session”, “Task”, “Run”).
- **Key:** a short string, typically a compression of the entity name, which uniquely identifies the entity when part of a filename (e.g. "sub", "ses", "task", "run").

- **Value type:** the requisite form of the value that gets specified alongside the key every time the entity appears in a filename. The value associated with an entity can be one of two possible types:
 - **Label:** an alphanumeric string, with the possibility of including the '+' character one or multiple times. (e.g. "scens01" is a valid label).
 - **Index:** a nonnegative integer, possibly prefixed with arbitrary number of 0s for consistent indentation (e.g. "1", "01" and "001" are all valid indexes).

The prescription of how an entity appears within any given filename is a string called **entity format**, while the specific manifestation of an entity within the name of a specific file is called **entity instance**. The entity format of an entity with key "key" can be either `key-<label>` or `key-<index>` depending on the value type of that entity. For example the entity format associated with "Session" is `ses-<label>`, while the entity format used for "Run" is `run-<index>`. By assigning a valid label or index in an entity format an entity instance is obtained. As an example, by substituting `<index>` in the entity format `run-<index>` with "1", the entity instance `run-1` is obtained. Entity instances are used in filenames to provide identifying information to the associated files. For example the previous obtained entity instance `run-1` can be used as part of a filename to identify its content as part of the data obtained from the first run done with a given set of acquisition parameters.

A filename in BIDS is constructed by composing a sequence of entities separated by the underscore character '_' (e.g. the entities `sub-01`, `ses-scens01` and `task-rest` can be concatenated as `sub-01_ses-scens01_task-rest`), and by terminating this sequence with a suffix, followed by a file extension. Those last two components of a filename in BIDS are defined as follows:

- **Suffix:** an alphanumeric string, located in the filename after all entities and following a final '_', immediately before the file extension. For example, in the filename `sub-01_ses-scens01_task-rest_run-1_recording-ecg_physio.tsv.gz`, the element `physio` is the file suffix, while the elements preceding it are entity instances. For some files the suffix varies depending on the modality associated to the data.

- **File extension:** the portion of the filename starting from the left-most period ('.') that is preceded by any other alphanumeric. For example, `.gitignore` does not have a file extension, whereas the file extension of `sub-01_ses-scans01_task-rest_run-1_recording-ecg_physio.tsv.gz` is `.tsv.gz`.

As a consequence of these naming rules, BIDS filenames do not contain whitespace characters and use only letters, numbers, hyphens, and underscores. This conventions facilitate automated processing through scripts and ensure compatibility across different operating systems by avoiding special characters. Furthermore, the BIDS standard requires to not rely on letter case (UPPERCASE and lowercase) to distinguish filenames, since some operating systems treat filenames in a case-insensitive manner.

In addition to defining rules for file naming, the BIDS standard also specifies how files must be organized within the dataset directory structure. According to this specification, a BIDS dataset is structured into four main hierarchical levels.

```
project/  
└─ subject/  
    └─ session/  
        └─ datatype/
```

The rules imposed by the standard for each one of this folder levels are the following:

- **Project level:** the folders at this level can have any name, the name assigned to a folder at this level should be descriptive of the referred dataset content. It is recommended to avoid spaces in the name, and use hyphens or underscores to separate words instead.
- **Subject level:** the folders at this level must be named using the format `sub-<participant label>`, there can be only one folder per participant and each folder must have an unique label.
- **Session level:** the folders at this level must be named using the format `ses-<session label>`, if there is no more than one session per participant this level can be omitted.

- **Datatype level:** the folders at this level represent functional groups of different kinds of data files. The data types folder names defined in BIDS for this level, and the functional group of data files they contain, are the following (listed in alphabetical order) [11, 8]:
 - **anat:** anatomical MRI data (structural imaging such as T1, T2, PD...)
 - **beh:** behavioral data
 - **dwi:** diffusion weighted MRI data
 - **eeg:** electroencephalography data
 - **emg:** electromyography
 - **fmap:** field inhomogeneity mapping data such as field maps
 - **func:** task based and resting state functional MRI data
 - **ieeg:** intracranial EEG data
 - **meg:** magnetoencephalography data
 - **micr:** microscopy data
 - **motion:** motion capture data
 - **mrs:** magnetic resonance spectroscopy data
 - **nirs:** near-infrared spectroscopy data
 - **perf:** arterial spin labeling data (perfusion data)
 - **pet:** positron emission tomography data
 - **phenotype:** measurement and survey data

A specific data type folder should be introduced in the dataset only if it is actually used, otherwise should be omitted. It is important to notice how data type is a concept different from the previously introduced concept of modality, but it is related to that. Different modalities are in fact related to different functional types of data (Table 4.1). It is also important to note that since data types are folders representing functional groups of different types of data, physiological data, like ECG recording, can be put in different datatype folders depending on the chosen modality.

	Modality							
	MRI	PET	meeg	behavioral	microscopy	NIRS	motion	MRS
Datatype	anat	pet	eeg	beh	micr	nirs	motion	mrs
	func		meg					
	dwi		ieeg					
	perf							

Table 4.1: Overview of BIDS datatypes folders across supported modalities [11]

Given a dataset that follows the hierarchical structure and naming rules described above, the presence of the file `dataset_description.json` at the root level of the directory hierarchy is required for the dataset to be considered BIDS-compliant. This file provides a machine-readable description of the dataset, and the fields it may contain are documented in detail in the BIDS specification [10]. Among these, only two fields are mandatory: **Name**, which specifies the name of the dataset, and **BIDSVersion**, which indicates the version of the BIDS specification used during dataset creation. The latter is particularly important for validating the dataset’s compliance with the standard. Some optional fields, namely **DatasetType**, **GeneratedBy**, and **SourceDatasets**, will be discussed in the following subsection (subsection 4.1.3) in the context of derivative data.

4.1.3 Source, raw and derivative data

Different types of data are produced in the full length of a study, those can be distinguished in source, raw and derivative data.

- **Source data:** data collected before harmonization, reconstruction, and/or file format conversion. These are the data as collected by the acquisition devices. The BIDS specification does not go into the details of recommending a particular naming scheme for this data, but recommend for it to follow BIDS naming convention where possible to ease the transition to a BIDS compliant dataset.
- **Raw data:** data unprocessed or minimally processed due to file format conversion. The BIDS standard was designed to describe, name and organize

to this kind of data. This data must be saved accordingly to the BIDS specification.

- **Derived data:** data obtained as final or partial results of the transformation of raw data during the analysis process. Data derived from a raw BIDS dataset can be BIDS compliant or not.

In order to separate correctly source, raw and derived data, the BIDS specification advises to store actual source files with the data instead of relying on links to external source repositories. This is done in order to maximize long term preservation, which would suffer if an external repository would not be available anymore.

At the same time, in order to protect the raw data from accidental changes, derivatives of the raw data must be kept separate from it. This makes easier to distinguish partial results from the raw data and share the latter. Retaining the source data is especially valuable in a case when conversion fails to preserve crucial metadata unique to specific acquisition setup.

Regarding the storage of derivatives, there are three ways they can be stored: The first two are the following:

- Inside a `derivatives/<pipeline-name>-<variant>` folder in the root of the BIDS dataset they have been derived from.
- As a standalone dataset independent from the BIDS dataset they have been derived from, with the option of keeping the original data inside the folder `sourcedata/` in the root of the dataset, and having the code use to generate them with the documentation in the `code/` and `docs/` folder.

If the derivatives are stored in this way they are considered a BIDS dataset, so they must include `dataset_description.json` at the root level of their dataset and they should adhere to the BIDS standard to the full extent possible. The third way to store derivatives is inside a “Study dataset”. A specific form of BIDS dataset created for housing inside it all the data produced during a study (source data, raw BIDS data and derivatives) described as follow:


```
study-1/
├── sourcedata/
│   ├── raw/
│   │   └── list of sourcedata datasets (BIDS or not)
├── derivatives/
│   └── list of <pipeline-name>-<variant> (BIDS or not)
└── dataset_description.json
```

Here only the `sourcedata/raw` directory must be a BIDS-compliant dataset, and the full structure is considered a fully compliant BIDS dataset.

When this convention is used it must be signaled inside the file `dataset_description.json` by setting the field `DatasetType` to `"study"`. It is also recommended to set `SourceDatasets` in `dataset_description.json` in the following way.

```
{
  "SourceDatasets": [ {"URL": "../sourcedata/raw/"} ]
}
```

This convention appears to be the most flexible out of the three available, for this reason it has been chosen for the multimodal dataset developed inside the laboratory ([section 4.2](#)).

It is important to notice that when storing the derivatives inside the `derivatives` folder using the format `<pipeline-name>-<variant>`, this same name should appear inside the `dataset_description.json` file in the field `GeneratedBy.Name`.

4.1.4 BIDS file format specification

While `sourcedata` and derived data do not have to perfectly adhere to the BIDS standard ([subsection 4.1.3](#)), raw datasets have to. Adhering to the standard is not just about following a specific dataset structure and naming, but it also imply following specific rules related to file formats. By following those rules only three types of data can be used inside a BIDS raw dataset:

- **NIfTI file format:** used for imaging data (`.nii.gz` recommended).
- **Tabular data:** in the form of `.tsv` files. Always present an header with column names in `snake_case`, with the exception of motion recording data

that do not use any header but have columns specified in another file (??). Here numbers must employ the dot '.' as decimal separator and character must be in UTF-8 encoding.

- **Compressed tabular data:** often in the form of `.tsv.gz` files. Without an header, their column names must be defined inside a sidecar JSON file in the `Columns` field.
- **Key-value files:** JSON (JavaScript Object Notation) files.

`.json` files are not only used for `dataset_description.json`, they can be associated to a TSV file, taking the name of JSON sidecars. In this case the JSON sidecar must have the same name (excluded the file extension) of the TSV file. When that happens the JSON file usually have one or more fields describing the column of the related TSV file, those fields have the same name as the column in the TSV file and they describe the column using a JSON object that can contains a specific set of fields that can be found in the appendix [Appendix A](#).

4.1.5 The Inheritance Principle

In some circumstances, there can be multiple data files for which all or a subset of the relevant metadata is precisely equivalent. In that situation it could be useful to define the metadata only once and place it in the filesystem in a way that it apply to all the relevant data files, and at the same time it does not apply to others data files. The set of rule determining which metadata is associated with which data file is called “Inheritance Principle”.

Further, because multiple metadata files may apply to an individual data file, the Principle defines the order of precedence of such metadata content; this is necessary for resolution of conflicts if the same metadata field contains different values in different metadata files

The rules are reported inside the BIDS specification documentation in great detail [9], for simplicity here will be reported only two example that will be useful for the later discussion of the dataset structure.

In the following dataset structures the inheritance principle is used to assign a single metadata file to multiple data files.

In [Figure 4.1](#) one single file is used to apply the metadata at all the others independently of their run element.

```
sub-01/
├── anat/
├── func/
│   ├── sub-01_task-xyz_acq-test1_run-1_bold.nii.gz
│   ├── sub-01_task-xyz_acq-test1_run-2_bold.nii.gz
│   └── sub-01_task-xyz_acq-test1_bold.json
```

Figure 4.1: Single metadata file applying to multiple data files.

In [Figure 4.2](#) the principles applied in the previous example ([Figure 4.1](#)) is generalized by only specifying the suffix of the files, and by applying the principle from an higher level in the directory structure.

```
sub-01/
├── anat/
├── func/
│   ├── sub-01_task-xyz_acq-test1_run-1_bold.nii.gz
│   ├── sub-01_task-xyz_acq-test1_run-2_bold.nii.gz
└── bold.json
```

Figure 4.2: Single metadata file applying to multiple data files from higher directory level.

4.1.6 Data Summary Files

Data summary files are modality-agnostic files defined in the BIDS standard. The files used for the purposes of this work are the following:

- `participants.tsv` and `participants.json`;
- `_scans.tsv` and `_scans.json`;
- `_sessions.tsv` and `_sessions.json`.

The file `participants.tsv` is used to describe properties of the participants, such as age and sex. If present in the dataset, it must include the column

`participant_id`, whose values follow the format `sub-<label>` and uniquely identify each participant. Each participant must be described by exactly one row in this file. The `participants.tsv` file must be located at a higher directory level than the participant directories.

When the participant is not human, the `species` column is recommended. If this column is not present, the participant is assumed to be `"homo sapiens"`.

The file `_scans.tsv` describes timing and other properties of each recording file within a session. Its columns typically include the required field `filename`, containing the relative path to the recording file within the session directory, and `acq_time`, which stores the timestamp corresponding to when the first data point of the recording was acquired. This file can be used for recording synchronization purposes.

The file `_sessions.tsv` describes variables that change between sessions for a specific participant. This file must include the column `session_id`, and each session must be described by exactly one row.

Similarly to `_scans.tsv`, this file may include an `acq_time` column. In this case, `acq_time` refers to the timestamp corresponding to when the first data point of the first run of that session was acquired.

The files `participants.json`, `_scans.json`, and `_sessions.json` are JSON sidecar files used to provide additional descriptions of the columns contained in the corresponding `.tsv` files.

4.2 Dog Behavior Study Dataset Structure

After introducing the general concepts of the BIDS standard in the previous section ([section 4.1](#)), this section presents the dataset structure designed for the multimodal dog behavior study.

```

DOG_BEHAVIOR_STUDY_DATASET/
├── dataset_description.json
├── README.md
├── CITATION.cff
├── LICENSE
├── CHANGES
├── code/
├── sourcedata/
│   └── source/
│       ├── participants.tsv
│       ├── participants.json
│       ├── sessions.json
│       ├── scans.json
│       └── sub-01/
│           ├── sub-01_sessions.tsv
│           └── ses-scens01/
│               ├── sub-01_ses-scens01_scans.tsv
│               ├── video/
│               │   ├── sub-01_ses-scens01_task-treat_run-1_recording-cam1_video.mp4
│               │   ├── sub-01_ses-scens01_task-treat_run-1_recording-cam2_video.mp4
│               │   ├── sub-01_ses-scens01_task-treat_run-1_recording-cam3_video.mp4
│               │   └── sub-01_ses-scens01_task-treat_run-1_recording-cam4_video.mp4
│               ├── audio/
│               │   ├── sub-01_ses-scens01_task-treat_run-1_recording-micdog_audio.wav
│               │   └── sub-01_ses-scens01_task-treat_run-1_recording-micowner_audio.wav
│               ├── physio/
│               │   ├── sub-01_ses-scens01_task-treat_run-1_recording-ecg_physio.tsv.gz
│               │   └── sub-01_ses-scens01_task-treat_run-1_recording-ecg_physio.json
│               └── motion/
│                   ├── sub-01_ses-scens01_task-treat_tracksys-mpu_run-1_motion.tsv
│                   ├── sub-01_ses-scens01_task-treat_tracksys-mpu_run-1_motion.json
│                   └── sub-01_ses-scens01_task-treat_tracksys-mpu_run-1_channels.tsv
└── derivatives/

```

Figure 4.3: Full BIDS structure for the Dog Behavior Study dataset

The dataset structure was designed following the principles of the BIDS standard while adapting it to the specific requirements of the dog behavior study. Since there are plans to extend the biosignal recordings in the future to include EEG data in addition to ECG, adopting a BIDS-inspired structure was considered a suitable

choice to ensure compatibility with existing neurophysiological data standards.

The dataset is organized as a *study dataset*. This allows laboratory members to store the dataset on the internal network and work on its contents while generating additional processed data inside the **derivatives** directory.

For each participant, two types of sessions are defined, as described in [chapter 3](#): *presessions* and *scenario sequences*. These sessions are identified using the prefixes **pres** and **scens**, respectively, followed by a zero-padded numerical identifier.

In standard BIDS datasets, physiological signals are typically stored directly within the modality directories. However, in this dataset ECG recordings may be used in combination with different types of behavioral or motion data during later analysis stages. For this reason, physiological signals are stored in a dedicated **physio** directory during the acquisition stage and may be reorganized into modality-specific structures when derivatives are generated.

Metadata files describing the dataset fields are provided in the appendix ([Appendix B](#)).

Software Tools for Dataset Creation

This chapter describes the collection of software tools developed during the traineeship to facilitate the creation of the multimodal dataset for dog behavior research at the TUAT Biosignal Informatics Laboratory.

The development of these tools followed a structured process, beginning with the requirements analysis ([section 5.1](#)). Based on these requirements, a data acquisition framework built on the Lab Streaming Layer (LSL) framework [[34](#)]([section 5.2](#)) and a set of additional tools for supporting the management of the dataset contents ([section 5.3](#)) were implemented using Python.

Together, these software tools enable the creation of the multimodal dog behavior dataset by facilitating the acquisition of experimental data and the integration of the corresponding metadata.

An example workflow illustrating how the different tools interact during the dataset creation process will be presented at the end of this chapter ([section 5.4](#)).

5.1 Requirements Analysis

The requirements for the realization of the software tools for dataset creation can be divided into two groups:

- **Functional requirements:** used to guide the development of the data acquisition framework.
- **Dataset management requirements:** used to guide the development of additional tools supporting the management of the dataset contents and the insertion of metadata.

Functional requirements were defined first. Based on the assumption that all recording devices are connected to computers located within the same local network, and taking into consideration the details of the previously discussed experimental setup ([chapter 3](#)), the following requirements were identified.

- The user must be able to add a participant to the dataset using a user-friendly graphical user interface (GUI). Each participant must correspond to an entry in the `participants.tsv` file of the dataset. Each participant entry must include a unique identifier, species (always set to `"canis familiaris"`), breed, sex, a field indicating whether the animal is neutered (boolean value), age expressed in months, weight expressed in kilograms, the number of hours per day spent with the owner, and a field specifying the provenance of the animal (`"pet store"`, `"private breeder"`, `"animal shelter"`, `"animal rescue organization"`, `"self bred"`, `"other owner"`, `"found as a stray"`, or `"other"`).

The unique identifier associated with each participant must follow the format `sub-<label>`, where `<label>` is a zero-padded integer. By default, the identifier is automatically assigned using the next available integer after the largest identifier already present in the dataset (e.g. `sub-08` will be assigned if `sub-07` is the highest existing identifier). However, the identifier can also be manually specified by the user if necessary.

- The user must be able to initiate a recording on all available devices connected to the computers in the laboratory network through a graphical in-

terface. Recordings must start at approximately the same time across all devices, with synchronization differences on the order of milliseconds.

To initiate a recording, the user must specify at least the identifier of the participant involved, the session type (“Presession” or “Scenario sequence”), and the task performed during the recording run. Session and run identifiers are automatically generated by default using the next available integer after the largest existing value (e.g., `ses-pres03` follows `ses-pres02` for the same participant, and `run-3` follows `run-2` within the same session). These identifiers may also be manually specified if required.

- Each computer in the laboratory must be able to record video streams from two cameras simultaneously. While recording or while waiting for the recording trigger, a live preview of the video streams must be displayed to the user. This feature allows researchers to adjust the position of the cameras in the experimental room without requiring the start of a recording.
- Each computer is connected to the receiver of a pair of wireless microphones. The acquisition software must be able to record of the audio signals received from these microphones.
- One computer is connected to the device measuring ECG and motion data (gyroscope and accelerometer). Each sample received from the device must be parsed in order to separate ECG data from motion data, and different sampling frequencies of the two signals must be handled correctly while recording the data.
- The actual start time of each recording must be reported in the corresponding `_scans.tsv` file associated with the recording session.
- All recorded data must be correctly named and saved in their appropriate location within the dataset according to the previously defined dataset structure ([chapter 4](#)).

Dataset management requirements were subsequently defined. Given the large volume of data recorded by the acquisition framework, streaming all data across the laboratory network during recording would not have been practical. For this

reason, it was decided to save the recorded data locally on each computer and later merge them into a single dataset.

Based on the need to merge locally recorded datasets and to add files to the dataset after the data have already been recorded, the following requirements were defined.

- It must be possible to merge datasets created on different computers without causing data duplication or loss.
- It must be possible to compute the content of the `_sessions.tsv` files and annotate the dataset with the JSON files required by the BIDS standard, in order to describe the content of the recorded data files and ensure reproducibility, after dataset creation.

5.2 Data Acquisition Framework

The data acquisition framework developed in the context of this thesis is a collection of Python programs designed to acquire data from a variety of recording devices and store the acquired data files, together with a minimal amount of meta-data (in the form of `_scans.tsv` files), according to the dataset structure defined in the previous chapter ([chapter 4](#)).

All this programs present a graphic user interface (GUI) for simplifying the interaction with the user.

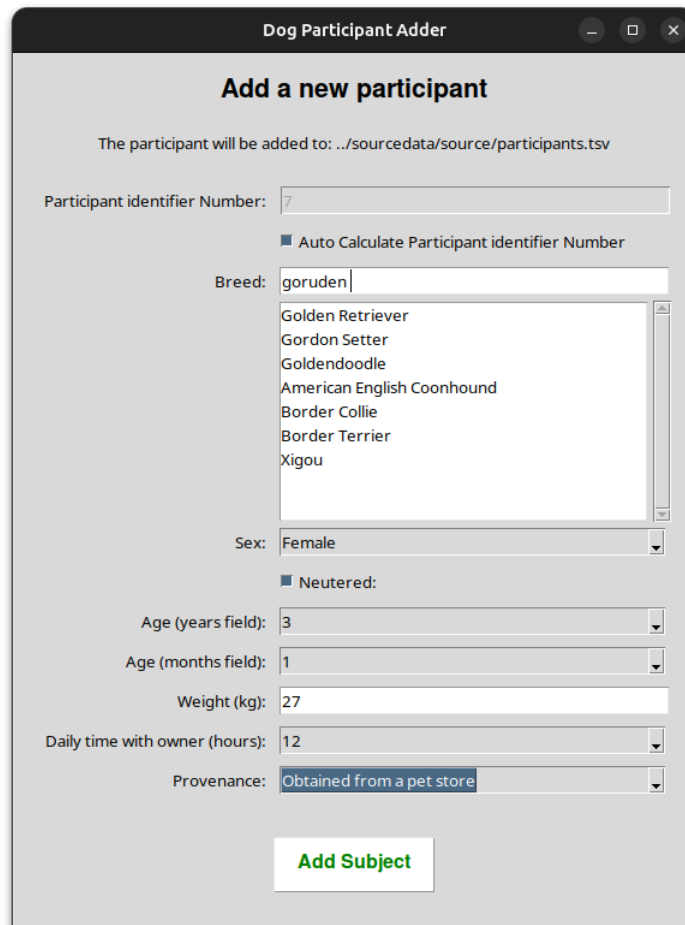
With the exception of **Dog Participant Adder** ([subsection 5.2.1](#)), all these programs operate using the Lab Streaming Layer (LSL) framework [34] for synchronizing the acquisition of data streams across multiple computers. In the following subsections, the role and functioning of each developed program will be described in detail in the following subsections.

All the developed Python code is available in the GitHub repository cited in the bibliography [13].

5.2.1 Dog Participant Adder

This program, whose filename is `participant_adder.py`, is an application developed in Python using the Tkinter package [22]. It provides a graphical user

interface (GUI) for adding a new participant entry to the `participants.tsv` file found inside the dataset. An image of the interface is shown in the following figure (Figure 5.1).



The screenshot shows a window titled "Dog Participant Adder" with a subtitle "Add a new participant". Below the subtitle, it states "The participant will be added to: ../sourcedata/source/participants.tsv". The form contains several fields: "Participant identifier Number" with a text input containing "7" and a checked checkbox "Auto Calculate Participant identifier Number"; "Breed" with a text input containing "goruden" and a dropdown menu showing a list of breeds (Golden Retriever, Gordon Setter, Goldendoodle, American English Coonhound, Border Collie, Border Terrier, Xigou); "Sex" with a dropdown menu showing "Female" and a checked checkbox "Neutered"; "Age (years field)" with a dropdown menu showing "3"; "Age (months field)" with a dropdown menu showing "1"; "Weight (kg)" with a text input containing "27"; "Daily time with owner (hours)" with a dropdown menu showing "12"; and "Provenance" with a dropdown menu showing "Obtained from a pet store". At the bottom, there is a green button labeled "Add Subject".

Figure 5.1: Dog Participant Adder application GUI

The “Participant Identifier Number” is automatically calculated by periodically performing a lookup in the `participants.tsv` file. However, the identifier can also be manually inserted by the user if necessary. If while writing on `participants.tsv` the file `participants.json` is not present, it gets automatically added to the dataset.

To avoid inconsistencies in the spelling of breed names in the “Breed” field, the program provides a list containing dog breeds compiled from Wikipedia [18]. The

list contains 650 different breeds, including common breeds, endangered breeds, crossbreeds, and a special entry named “Mixed Breed” to cover cases where the breed cannot be precisely identified.

To simplify the search for a specific breed, the list is dynamically updated while the user types in the “Breed” field using a tiered sorting strategy. In particular, breeds are ordered according to the following criteria:

- breeds that perfectly match the beginning of the typed string;
- breeds matching the beginning of the string after converting the breed name to plain ASCII characters;
- the remaining breeds ordered based on a string similarity score computed using the `fuzz.WRatio` function from the RapidFuzz Python package [21].

This approach helps prevent typographical errors and makes it easier to search for breeds when the exact spelling is uncertain.

It is important to note that, for ease of use, the age of the participant dog in the interface is entered by specifying years and months. However, the corresponding entry in the `participants.tsv` file is automatically converted and stored as the total number of months, following the practice adopted by the veterinary laboratory.

5.2.2 LSL Recording Commander

This program, whose filename is `commander.py`, is an application developed in Python using the Tkinter package [22]. It provides a graphical user interface (GUI) used to broadcast recording trigger messages on the local network using the Lab Streaming Layer (LSL) framework [34].

Given the large volume of data involved in the experiment, LSL is used only for sending control messages for starting and stopping recordings at extremely low latency, rather than for transmitting the recorded data streams themselves.

An image of the application interface is shown in the following figure (Figure 5.2).

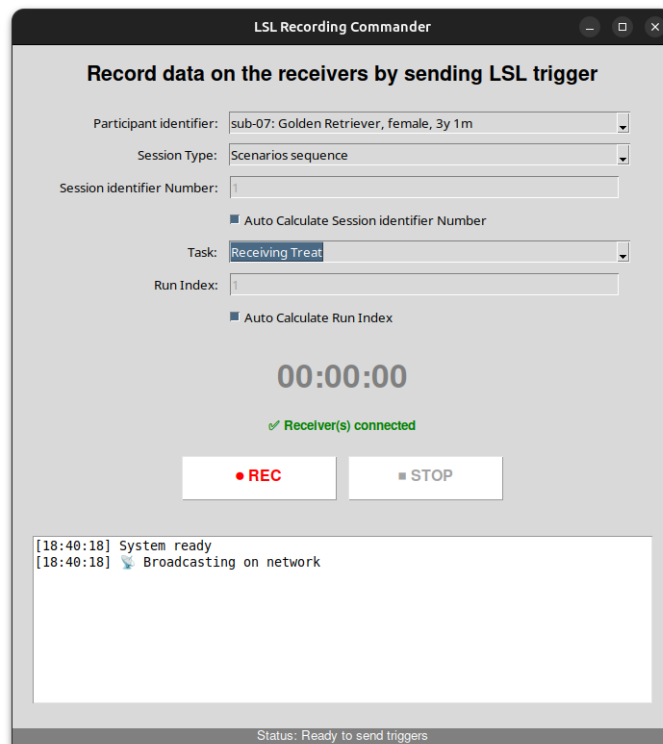


Figure 5.2: LSL Recording Commander application GUI

The development of this software was started by Puwadej Leelasiri, one of the doctoral student at the laboratory, as single application for sending a start recording signal to other devices in the LSL local network. During my traineeship i expanded on that concept integrating the application with the underlying dataset for displaying the participant in the UI, calculating the “Session identifier Number” and “Run Index” and increased the complexity of the start recording trigger signal for better integration of all the application in the data acquisition framework.

The field “Participant identifier” allows the user to select the participant involved in the recording session. The interface displays additional information extracted from each participant entry in `participants.tsv`, making it easier to identify the correct participant without relying only on the identifier string.

While the system is not recording, the program periodically performs a lookup in the `participants.tsv` file in order to always display an updated list of participants available for selection and their details.

The value of the “Session identifier Number” field is automatically calculated

once the “Session Type” is selected. The program determines the next available index by searching the dataset for sessions of the same type involving the selected participant. Similarly, the “Run Index” value is automatically calculated once both “Session Type” and “Task” are selected by searching for previously recorded runs within the same session and assigning the next available index. Both the “Session identifier Number” and the “Run Index” can also be manually specified by the user if necessary.

When the “REC” button is pressed, the input fields of the interface become disabled and a start recording trigger message is broadcast on the local network using LSL. This message instructs the other components of the data acquisition framework to start recording and gives them the information necessary to saving the recorded data in the appropriate directories and using the correct filename elements. At the same time, a timer indicating the duration of the recording is started, the lower portion of the interface turns red to signal that recording is in progress, and the “STOP” button becomes interactable (Figure 5.3).

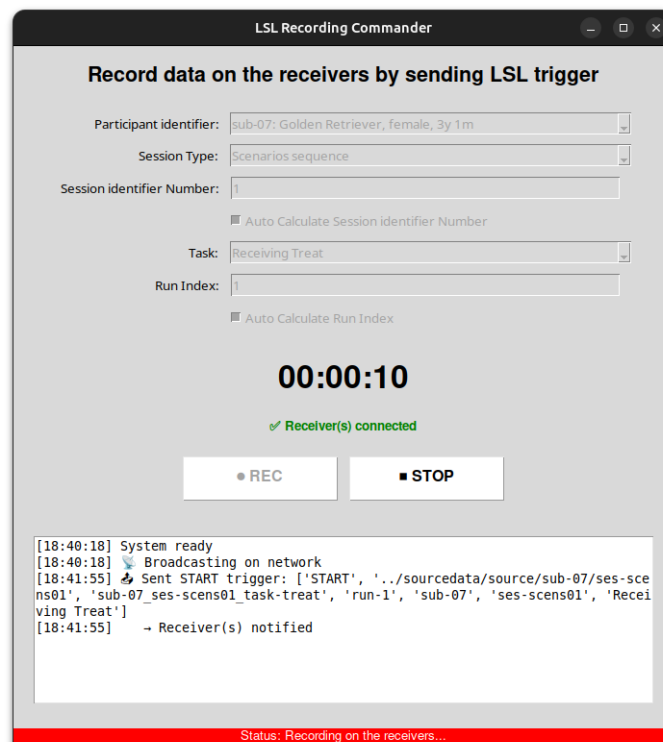


Figure 5.3: LSL Recording Commander application GUI during recording

When the “STOP” button is pressed, a stop recording trigger message is broadcast on the local network. This message instructs the other components of the data acquisition framework to stop recording data and to write the timestamp corresponding to the start of each recorded file into the appropriate `_scans.tsv` files. After the stop command is issued, the program returns to its initial state, with the input fields enabled and the timer reset (Figure 5.4).

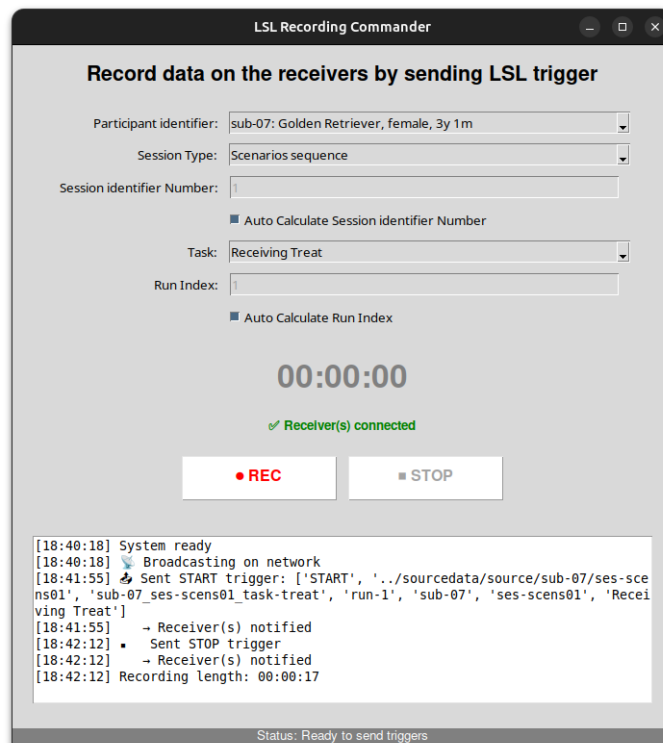


Figure 5.4: LSL Recording Commander application GUI after stopping the recording

The trigger message broadcast by the commander, both when starting and stopping a recording, consists of a list of seven strings. The first element of the list, `trigger[0]`, contains either the string "START" or "STOP", indicating whether the receivers should start or stop recording.

If the message begins with "STOP", all the remaining fields are empty. If it begins with "START", the remaining six fields contain information used by the receiving programs to correctly store and name the recorded data and related

`_scans.tsv` metadata.

- `trigger[1]`: path where the files of the session must be saved.
- `trigger[2]`: filename prefix common to all the data recorded during the session.
- `trigger[3]`: the element `run-<index>` to be included in the filenames of the recorded data before the suffix.
- `trigger[4]`: participant identifier (format `sub-<label>`) used to determine the filenames of the `_sessions.tsv` and `_scans.tsv` files.
- `trigger[5]`: session identifier (format `ses-<label>`) used to determine the filenames of the `_scans.tsv` files.
- `trigger[6]`: currently unused. It is reserved for future extensions, such as filling JSON sidecar files directly during recording.

The advantage of this design is that it allows the data acquisition framework to be easily extended in the future. Additional programs supporting new recording devices can be integrated into the system simply by implementing the ability to correctly interpret the start and stop trigger messages broadcast by the commander.

5.2.3 LSL Video and Audio Recorder

This program, whose filename is `recorder_audio_video.py`, is an application developed in Python using the Tkinter package [22]. It provides a graphical user interface (GUI) used to display a live preview of the two cameras connected to the computer and includes a log box reporting the reception of LSL trigger messages and the operations performed by the program.

As suggested by its name, the application main function is recording the audio data captured by the wireless microphones connected to the computer, and the video data captured by the two cameras connected to the computer. Audio and video recording start simultaneously once the LSL start recording trigger signal is received from the commander.

An image of the application interface is shown in the following figure (Figure 5.5).

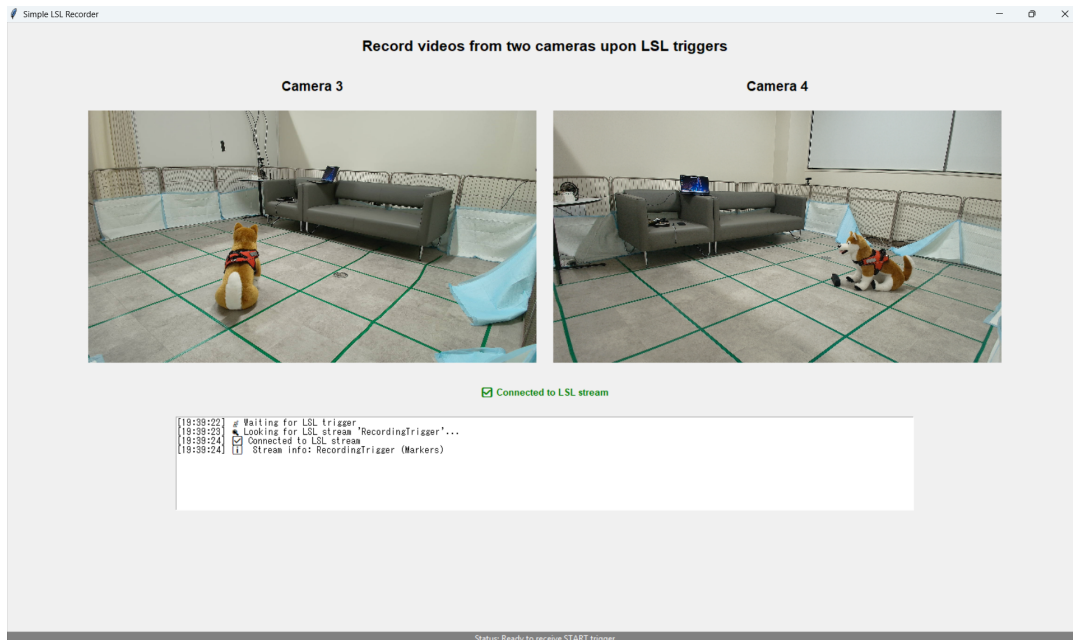


Figure 5.5: LSL Video and Audio Recorder application GUI

Like the “LSL Recording Commander”, the development of this software was started by the doctoral student Puwadej Leelasiri. Initially it was developed as a software for video recording only without GUI, my work proceeded from there introducing the GUI components, automatic connection to LSL, support to wireless microphones and metadata files creation.

In order to continuously display an updated video preview while avoiding frame loss during recording, the application uses two separate frame queues for each camera. When a frame is captured from the camera, it is inserted into both queues.

The first queue, used for the preview display, has a capacity of a single frame. When the preview image is rendered on the screen, the queue is emptied, allowing it to immediately accept the next frame. This ensures that the preview always shows the most recent frame captured by the camera.

The second queue is used for recording and can store up to 1000 frame, this helps absorb temporary delays in disk writing operations, preventing frame drops

when the recording system experiences short bursts of high I/O load. Frames stored in this queue are written to disk asynchronously in the background while the recording is in progress. This mechanism allows the preview to remain responsive while ensuring that all frames are saved without loss during recording.

5.2.4 LSL ECG and MPU Recorder

This program, whose filename is `recorder_ecg_mpu.py`, is an application developed in Python using the PyQt5 package [7]. It provides a graphical user interface (GUI) used to connect via Bluetooth to the custom device employed for measuring ECG and motion data. The interface also displays a real-time graphical preview of the signals received from the device.

As suggested by its name, the primary purpose of the application is to record the ECG and motion data captured by the device connected to the computer.

An image of the application interface is shown in the following figure (Figure 5.6).



Figure 5.6: LSL ECG and MPU Recorder application GUI

This software was mainly developed by Motoi Noda, the laboratory member

who designed the custom sensing device used in the experiments ([chapter 3](#)). My contribution to the code consisted of integrating the application with the rest of the data acquisition framework. In particular, I implemented the functionality required for interpreting the LSL trigger messages and for saving the recorded signals in separate BIDS-compatible files taking their original sampling rate in consideration.

The signal transmitted by the device contains both ECG and motion data. Therefore, the software parses the incoming signal to separate the two data streams while correctly handling the different sampling frequencies of the sensors integrated in the device.

As described in [chapter 3](#), the ECG signal is sampled at 300 Hz, while the motion data produced by the MPU sensor are sampled at 100 Hz. Because the data are transmitted together at the higher sampling rate, each motion value is repeated three times in the received signal. To avoid incorrectly treating these repeated values as distinct measurements, the acquisition software resamples the motion data by selecting only one value every three samples before saving the data to disk.

5.3 Dataset Management Tools

The dataset management tools are a collection of Python programs developed to support the management of the dataset contents and the insertion of additional metadata, satisfying the dataset management requirements described in [section 5.1](#).

Unlike the programs described in the previous section ([section 5.2](#)), these tools do not provide a graphical user interface. Instead, they are designed to be used from the command line after the data recording process has been completed.

5.3.1 Dataset Merging Tool

This program, whose filename is `safely_merge_datasets.py`, is a command-line tool designed to merge multiple datasets into a single destination dataset without risking data loss.

This tool was developed as a consequence of the decision to store recorded data locally on each computer used during the experiments discussed in [chapter 3](#).

Although the data acquisition framework ensures that all recorded files are saved in the correct position within the dataset structure, the resulting data remain distributed across multiple computers. The dataset merging tool automates the process of combining these fragmented datasets into a single unified dataset, ensuring that datasets recorded across multiple machines can be safely consolidated without manual intervention.

The tool is used by invoking the program name followed by a list of dataset root directories. The contents of all specified datasets are merged into the last directory in the list, which is used as the destination dataset. If the destination directory does not already exist, it is automatically created and populated with the contents of the source datasets.

For example:

```
python safely_merge_datasets.py datasetA datasetB datasetC
```

will merge the contents of `datasetA` and `datasetB` into `datasetC` while avoiding conflict between the content of the three datasets.

The merging process is designed to be safe. If a file with the same name as the one being copied already exists in the destination dataset, the file is skipped in order to prevent overwriting existing data. However, if the file is a `.tsv` file, a different strategy is applied: the entries present in the incoming file but missing in the destination file are appended to the existing file instead of being ignored.

This special handling is necessary because the only files expected to have identical names but different contents in datasets generated by the proposed data acquisition framework are the `_scans.tsv` files. These files contain the timestamps of the recordings and are therefore essential for signal synchronization. They are also used to compute the `acq_time` column in the `_sessions.tsv` metadata files and the `StartTime` fields in the JSON sidecar files for physiological data.

During the merging process, all operations performed by the tool (such as copying files, merging TSV entries, or skipping files) are printed to the terminal using color-coded messages. At the end of the process, a summary is displayed showing how many times each operation was performed.

By using the `-verbose` option, the tool can also display detailed information about the merge process, including the individual TSV entries that were added or ignored.

An example of the execution of this tool and the corresponding output is shown

in Figure 5.7.

```

• (venv) giovanni@giovanni-HP-Laptop:~/Desktop/dataset_creation_tools$ python3 merge_datasets.py DOG_BEHAVIOR_DATASET_1 DOG_BEHAVIOR_DATASET_2 DOG_BEHAVIOR_DATASET_3 DOG_BEHAVIOR_DATASET_4
T.4 --verbose
--- MERGE START ---
Cpd: DOG BEHAVIOR DATASET 1/participants.tsv
Cpd: DOG BEHAVIOR DATASET 1/sub-01/ses-rest01/scans.tsv
Cpd: DOG BEHAVIOR DATASET 1/sub-01/ses-rest01/beh/sub-01 ses-rest01 task-rest recording-cam3 video.mp4
Cpd: DOG BEHAVIOR DATASET 1/sub-01/ses-rest01/beh/sub-01 ses-rest01 task-rest recording-cam4 video.mp4
Cpd: DOG BEHAVIOR DATASET 1/sub-01/ses-rest01/beh/sub-01 ses-rest01 task-rest recording-ownermic audio.wav
Cpd: DOG BEHAVIOR DATASET 1/sub-01/ses-pres01/scans.tsv
Cpd: DOG BEHAVIOR DATASET 1/sub-01/ses-pres01/beh/sub-01 ses-pres01 task-freebeh recording-ownermic audio.wav
Cpd: DOG BEHAVIOR DATASET 1/sub-01/ses-pres01/beh/sub-01 ses-pres01 task-freebeh recording-cam4 video.mp4
Cpd: DOG BEHAVIOR DATASET 1/sub-01/ses-pres01/beh/sub-01 ses-pres01 task-freebeh recording-cam3 video.mp4
Mln: DOG BEHAVIOR DATASET 2/sub-01/ses-rest01/scans.tsv, line: beh/sub-01 ses-rest01 task-rest recording-cam1 video.mp4 2026-01-29T10:44:30.930150
Mln: DOG BEHAVIOR DATASET 2/sub-01/ses-rest01/scans.tsv, line: beh/sub-01 ses-rest01 task-rest recording-cam2 video.mp4 2026-01-29T10:44:30.930733
Cpd: DOG BEHAVIOR DATASET 2/sub-01/ses-rest01/beh/sub-01 ses-rest01 task-rest recording-dogmic audio.wav 2026-01-29T10:44:30.932576
Cpd: DOG BEHAVIOR DATASET 2/sub-01/ses-rest01/beh/sub-01 ses-rest01 task-rest recording-cam2 video.mp4
Cpd: DOG BEHAVIOR DATASET 2/sub-01/ses-rest01/beh/sub-01 ses-rest01 task-rest recording-cam1 video.mp4
Mln: DOG BEHAVIOR DATASET 2/sub-01/ses-pres01/scans.tsv, line: beh/sub-01 ses-pres01 task-freebeh recording-cam1 video.mp4 2026-01-29T10:44:01.887593
Mln: DOG BEHAVIOR DATASET 2/sub-01/ses-pres01/scans.tsv, line: beh/sub-01 ses-pres01 task-freebeh recording-cam2 video.mp4 2026-01-29T10:44:01.888633
Cpd: DOG BEHAVIOR DATASET 2/sub-01/ses-pres01/beh/sub-01 ses-pres01 task-freebeh recording-dogmic audio.wav 2026-01-29T10:44:01.890692
Cpd: DOG BEHAVIOR DATASET 2/sub-01/ses-pres01/beh/sub-01 ses-pres01 task-freebeh recording-cam1 video.mp4
Mln: DOG BEHAVIOR DATASET 3/sub-01/ses-rest01/scans.tsv, line: beh/sub-01 ses-rest01 task-rest acq-ecg+gyro beh.tsv 2026-01-29T10:44:30.942576
Cpd: DOG BEHAVIOR DATASET 3/sub-01/ses-rest01/beh/sub-01 ses-rest01 task-rest acq-ecg+gyro beh.tsv
Mln: DOG BEHAVIOR DATASET 3/sub-01/ses-pres01/scans.tsv, line: beh/sub-01 ses-pres01 task-freebeh acq-ecg+gyro beh.tsv 2026-01-29T10:44:01.990692
Cpd: DOG BEHAVIOR DATASET 3/sub-01/ses-pres01/beh/sub-01 ses-pres01 task-freebeh acq-ecg+gyro beh.tsv
--- MERGE END ---

Input datasets: ['DOG_BEHAVIOR_DATASET_1', 'DOG_BEHAVIOR_DATASET_2', 'DOG_BEHAVIOR_DATASET_3']
Output dataset: DOG_BEHAVIOR_DATASET_4
Datasets merge completed:
18 files copied
8 tsv lines merged
0 tsv lines skipped
0 files ignored
0 errors

```

Figure 5.7: Example output of the dataset merging tool

5.3.2 Annotate Dataset

This tool consists of a wrapper script that invokes the function

`annotate_dataset(dataset_root: Path),`

implemented in the file `dataset_utils.py`. This module is used by multiple components of the data acquisition framework and contains utility functions frequently required by the different programs, reducing code duplication and centralizing logic related to dataset management.

Given a dataset following the structure described in the previous chapter ([chapter 4](#)), this function annotates the dataset by automatically generating the meta-data files required for proper dataset documentation. This step ensures that the dataset is fully compliant with the BIDS metadata requirements after the recording and merging processes have been completed, so that when in the future a raw dataset will be created starting from source those files will be already prepared.

In particular, the following files are created:

- `scans.json` in the dataset root directory;

- `sub-<label>_sessions.tsv` inside each participant directory;
- `sessions.json` in the dataset root directory;
- a JSON sidecar file for each `_recording-ecg_physio.tsv.gz` file;
- a JSON sidecar file and a `channels.tsv` file for each `motion.tsv` file located in the `motion` directory.

5.4 Example Workflow: Recording a First Session

To better illustrate how the different tools of the data acquisition framework interact during the creation of the multimodal dataset, this section describes a typical workflow for recording the first experimental session of a new participant. Since the project is still in its early stages, it is common for dog owners to visit the laboratory with their dogs for the first time.

When the owner and the dog arrive at the laboratory, they are first guided by the researchers to the experimental room. At this stage, the researchers collect the relevant information about the dog (such as breed, age, and other metadata). Using this information, the dog is added to the `participants.tsv` file through the “Dog Participant Adder” application ([subsection 5.2.1](#)).

Once the dog has been added to the list of participants, it becomes possible to begin recording a pre-session. This is done using the “LSL Recording Commander” application ([subsection 5.2.2](#)) together with the other programs of the data acquisition framework. In this initial phase, the veterinarian may advise against recording ECG data if the dog shows signs of discomfort. In such cases, only the “LSL Video and Audio Recorder” applications ([subsection 5.2.3](#)) are started on the two laboratory computers in order to record only video data. Using the “LSL Recording Commander”, the first recording run of the pre-session is started simultaneously on both machines.

After approximately twenty minutes, during which the dog receives treats and becomes familiar with the laboratory environment and the researchers, the dog usually appears calmer according to both the owner and the veterinarian. At this point, the veterinarian may decide that it is possible to attempt fitting the dog with the special bib containing the ECG and motion sensors.

With the assistance of the owner, the bib is placed on the dog and the sensing device is turned on. The bib also provides space for attaching the wireless microphone, allowing audio data to be recorded as well. At this stage, the initial recording is stopped and the collected data are automatically saved in the correct position within the dataset structure. The “LSL Recording Commander” automatically updates the run index to 2 (while the session identifier remains 1), preparing the system for the next recording run.

The “LSL ECG and MPU Recorder” application is then opened and connected via LSL to the commander, resulting in three different recording applications running across the two computers. A new recording is then started from the commander, this time with run index 2, allowing the acquisition of ECG, motion, audio, and video data simultaneously.

After approximately thirty minutes of recording, the session concludes and the recording is stopped.

Once the owner and the dog leave the laboratory, the recorded dataset remains distributed across the two computers used during the session. The `sourcedata` dataset directory from each computer is therefore copied to an external storage device and transferred to the biosignal laboratory. There, the “Dataset Merging Tool” ([subsection 5.3.1](#)) is used to combine the datasets from the two machines into a single unified dataset while preserving the contents of the `_scans.tsv` files. Finally, the merged dataset can be enriched with additional metadata by running the “Annotate Dataset” script ([subsection 5.3.2](#)).

Conclusion

6.1 Summary of Contributions

During the traineeship at the TUAT Biosignal Informatics Laboratory, my work focused on designing the structure of a multimodal dataset for dog behavior research, and on developing a data acquisition framework capable of recording data from multiple sensors and capable to automatically store the collected data, together with the associated metadata, in the correct position within the defined dataset structure.

The first contribution of this work was the design of a dataset structure for storing source data using a BIDS-inspired approach. This design prepares the laboratory for the future creation of a fully BIDS-compliant raw dataset and supports potential extensions of the experimental setup, such as the inclusion of EEG recordings from dogs, thanks to the precise set of requirements of the BIDS specification.

The second major contribution was the development of a flexible data acquisition framework. When the traineeship began, the system was designed to only record video data in a synchronous way in a previously defined folder and without any GUI. During the development process, the system evolved into a collection

of applications capable of interacting with multiple sensors while coordinating the acquisition process and the storage of data within the dataset structure. Furthermore the resulting framework was designed to remain extensible and adaptable to future changes in the laboratory’s experimental needs.

The work carried out during this project brought several advantages to the laboratory operations:

- **Creation of a shared vocabulary:** The adoption of standardized terms such as participant, session, run, task, and recorder helped to establish a consistent internal vocabulary for describing experiments and datasets. This improved communication among researchers and simplified data identification.
- **Standardized data generation:** Clear procedures for data acquisition were established, including the automatic generation of useful metadata, such as timestamps stored in the `_scans.tsv` files for checking the synchronization of the recordings. This reduces the likelihood of user errors and improves the reliability of the recorded data.
- **Support for multiple research projects:** The dataset structure evolved through several iterations before reaching its current form. Each revision aimed to improve the usability of the dataset for researchers and to create a structure capable of supporting multiple studies without being tailored to a single experiment.

6.2 Future Developments

Several future enhancements can further improve the system developed during this work:

- **Improving usability of “LSL Video and Audio Recorder”:** The current implementation of the application only provides a preview of the video streams and does not display the signals captured by the microphones. Adding a visualization of the audio signals would allow researchers to visually monitor the quality of the microphone placement and verify that

the recorded audio is being captured correctly. Additionally, introducing a graphical option to select the input devices would improve usability, as the current implementation requires modifying a global variable in the source code to change the recording devices.

- **Integration of additional sensors:** The current data acquisition framework was designed to be extensible. Future work may include integrating additional sensors into the experimental setup by developing new applications compatible with the existing framework.
- **Improved data storage infrastructure:** Currently, data recorded during experiments are stored locally on the laboratory computers and later merged manually. Introducing a centralized storage system, such as a Network Attached Storage (NAS), would simplify data management and reduce the need to transfer datasets using external hard drives.

*Sidecar JSON Fields for a TSV
File Column*

Key name	Requirement	Level	Data type	Description
LongName	OPTIONAL		string	Long (unabbreviated) name of the column.
Description	RECOMMENDED		string	Free-form natural language description. The description of the column.
Format	OPTIONAL		string	Permitted formats for values in the described column. For a list of valid values for this field, see the associated glossary entry.
Levels	RECOMMENDED		object	For categorical variables: An object of possible values (keys) and their descriptions (values).
Units	RECOMMENDED		string	Measurement units for the associated variable. SI units in CMIXF formatting are RECOMMENDED (see Units).
Delimiter	OPTIONAL		string	If rows in a column may be interpreted as a list of values, the character that separates one value from the next.
TermURL	RECOMMENDED		string	URL pointing to a formal definition of this type of data in an ontology available on the web. For example: https://www.ncbi.nlm.nih.gov/mesh/68008297 for "male".
HED	OPTIONAL		string or object of strings	Hierarchical Event Descriptor (HED) information, see the HED Appendix for details.
Maximum	OPTIONAL		number	Maximum value a column entry is permitted to have.
Minimum	OPTIONAL		number	Minimum value a column entry is permitted to have.

Table A.1: Metadata fields and requirements for tabular data

Multimodal Dataset Metadata Files

B.1 dataset_description.json

```
1 {
2     "Name": "Dog Behavior Multimodal Study Dataset",
3     "BIDSVersion": "1.10.1",
4     "DatasetType": "study",
5     "Keywords": [
6         "dog behavior",
7         "animal behavior",
8         "multimodal data"
9     ]
10 }
```

B.2 CHANGES

```
1 1.0.0 2026-03-03
2 - First official presentation of the dataset structure and data
   ↪ acquisition framework.
```

B.3 participants.tsv

```

1 participant_id      species      breed      sex      neutered
   ↪      age_months      weight_kg      daily_time_with_owner_hou
   ↪ rs      provenance
2 sub-01      canis familiaris      Shiba Inu      male      True
   ↪      78      30.0      20      pet store
3 sub-02      canis familiaris      Sabueso Español      male
   ↪      True      88      30.0      10      pet store
4 sub-03      canis familiaris      Golden Retriever      female
   ↪      True      37      27.0      12      pet store

```

B.4 participants.json

```

1 {
2   "participant_id": {
3     "LongName": "Participant identifier",
4     "Description": "Unique participant identifier",
5     "Format": "string"
6   },
7   "species": {
8     "LongName": "Species",
9     "Description": "Biological species of the participant",
10    "Format": "string",
11    "Levels": {
12      "canis familiaris": "Domestic dog"
13    }
14  },
15  "breed": {
16    "LongName": "Breed",
17    "Description": "Dog breed as reported by the breeder or shelter",
18    "Format": "string",
19    "Levels": {
20      "Affenpinscher": "Affenpinscher",
21      "Afghan Hound": "Afghan Hound",

```



```
22     "African hairless dog (Abyssinian sand terrier)": "African
    ↪ hairless dog (Abyssinian sand terrier)",
23     "Africanis": "Africanis",
24     "Aidi": "Aidi",
25     "Airedale Terrier": "Airedale Terrier",
26     "Akbash": "Akbash",
27     "Akita": "Akita",
28     "Aksaray Malaklisi": "Aksaray Malaklisi",
29     "Alano Español": "Alano Español",
30     "Alapaha Blue Blood Bulldog": "Alapaha Blue Blood Bulldog",
31     "Alaskan Husky": "Alaskan Husky",
32     "Alaskan Klee Kai": "Alaskan Klee Kai",
33     "Alaskan Malamute": "Alaskan Malamute",
34     "Alaunt": "Alaunt",
35     "Alopekis": "Alopekis",
36     "Alpine Dachsbracke": "Alpine Dachsbracke",
37     "Alpine Mastiff": "Alpine Mastiff",
38     "American Bulldog": "American Bulldog",
39     "American Bully": "American Bully",
40     "American Cocker Spaniel": "American Cocker Spaniel",
41     "American English Coonhound": "American English Coonhound",
42     "American Eskimo Dog": "American Eskimo Dog",
43     "American Foxhound": "American Foxhound",
44     "American Hairless Terrier": "American Hairless Terrier",
45     "American Leopard Hound": "American Leopard Hound",
46     "American Pit Bull Terrier": "American Pit Bull Terrier",
47     "American Staffordshire Terrier": "American Staffordshire
    ↪ Terrier",
48     "American Staghound": "American Staghound",
49     "American Water Spaniel": "American Water Spaniel",
50     "Andalusian Terrier": "Andalusian Terrier",
51     "Anglo-Français de Petite Vénérerie": "Anglo-Français de Petite
    ↪ Vénérerie",
52     "Appenzeller Sennenhund": "Appenzeller Sennenhund",
53     "Argentine Pila": "Argentine Pila",
54     "Argentine Polar Dog": "Argentine Polar Dog",
55     "Ariège Pointer": "Ariège Pointer",
```

```
56     "Ariégeois": "Ariégeois",
57     "Armant": "Armant",
58     "Armenian Gampr": "Armenian Gampr",
59     "Artois Hound": "Artois Hound",
60     "Assyrian Mastiff": "Assyrian Mastiff",
61     "Australian Cattle Dog": "Australian Cattle Dog",
62     "Australian Kelpie": "Australian Kelpie",
63     "Australian Shepherd": "Australian Shepherd",
64     "Australian Silky Terrier": "Australian Silky Terrier",
65     "Australian Stumpy Tail Cattle Dog": "Australian Stumpy Tail
    ↪ Cattle Dog",
66     "Australian Terrier": "Australian Terrier",
67     "Austrian Black and Tan Hound": "Austrian Black and Tan
    ↪ Hound",
68     "Austrian Pinscher": "Austrian Pinscher",
69     "Azawakh": "Azawakh",
70     "Bc Hà": "Bc Hà",
71     "Bakharwal": "Bakharwal",
72     "Banjara Hound (Vanjari Hound)": "Banjara Hound (Vanjari
    ↪ Hound)",
73     "Bankhar Dog": "Bankhar Dog",
74     "Barak Hound": "Barak Hound",
75     "Barbado da Terceira": "Barbado da Terceira",
76     "Barbet": "Barbet",
77     "Basenji": "Basenji",
78     "Basque Shepherd Dog": "Basque Shepherd Dog",
79     "Basset Artésien Normand": "Basset Artésien Normand",
80     "Basset Bleu de Gascogne": "Basset Bleu de Gascogne",
81     "Basset Fauve de Bretagne": "Basset Fauve de Bretagne",
82     "Basset Hound": "Basset Hound",
83     "Bavarian Mountain Hound": "Bavarian Mountain Hound",
84     "Beagle": "Beagle",
85     "Beagle-Harrier": "Beagle-Harrier",
86     "Beaglier": "Beaglier",
87     "Bearded Collie": "Bearded Collie",
88     "Beauceron": "Beauceron",
89     "Bedlington Terrier": "Bedlington Terrier",
```

```
90     "Belgian Mastiff": "Belgian Mastiff",
91     "Belgian Shepherd": "Belgian Shepherd",
92     "Bergamasco Shepherd": "Bergamasco Shepherd",
93     "Berger Picard": "Berger Picard",
94     "Bernedoodle": "Bernedoodle",
95     "Bernese Mountain Dog": "Bernese Mountain Dog",
96     "Bichon Frisé": "Bichon Frisé",
97     "Biewer Terrier": "Biewer Terrier",
98     "Billy": "Billy",
99     "Black and Tan Coonhound": "Black and Tan Coonhound",
100    "Black and Tan Terrier": "Black and Tan Terrier",
101    "Black Mouth Cur": "Black Mouth Cur",
102    "Black Norwegian Elkhound": "Black Norwegian Elkhound",
103    "Black Russian Terrier": "Black Russian Terrier",
104    "Bloodhound": "Bloodhound",
105    "Blue Lacy": "Blue Lacy",
106    "Blue Picardy Spaniel": "Blue Picardy Spaniel",
107    "Bluetick Coonhound": "Bluetick Coonhound",
108    "Boerboel": "Boerboel",
109    "Bohemian Shepherd": "Bohemian Shepherd",
110    "Bohemian Spotted Dog": "Bohemian Spotted Dog",
111    "Bolognese": "Bolognese",
112    "Border Collie": "Border Collie",
113    "Border Terrier": "Border Terrier",
114    "Borzoi": "Borzoi",
115    "Boston Terrier": "Boston Terrier",
116    "Bouvier des Ardennes": "Bouvier des Ardennes",
117    "Bouvier des Flandres": "Bouvier des Flandres",
118    "Boxer": "Boxer",
119    "Boykin Spaniel": "Boykin Spaniel",
120    "Bracco Italiano": "Bracco Italiano",
121    "Braque d'Auvergne": "Braque d'Auvergne",
122    "Braque du Bourbonnais": "Braque du Bourbonnais",
123    "Braque Dupuy": "Braque Dupuy",
124    "Braque Français": "Braque Français",
125    "Braque Saint-Germain": "Braque Saint-Germain",
126    "Brazilian Terrier": "Brazilian Terrier",
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127 "Briard": "Briard",
128 "Briquet de Provence": "Briquet de Provence",
129 "Briquet Griffon Vendéen": "Briquet Griffon Vendéen",
130 "Brittany": "Brittany",
131 "Broholmer": "Broholmer",
132 "Bruno Jura Hound": "Bruno Jura Hound",
133 "Buckhound": "Buckhound",
134 "Bucovina Shepherd Dog": "Bucovina Shepherd Dog",
135 "Bulgarian Hound": "Bulgarian Hound",
136 "Bulgarian Scenthound": "Bulgarian Scenthound",
137 "Bull Arab": "Bull Arab",
138 "Bull Terrier": "Bull Terrier",
139 "Bulldog": "Bulldog",
140 "Bullenbeisser": "Bullenbeisser",
141 "Bullmastiff": "Bullmastiff",
142 "Bully Kutta": "Bully Kutta",
143 "Burgos Pointer": "Burgos Pointer",
144 "Ca de Bou": "Ca de Bou",
145 "Ca Mè Mallorquí": "Ca Mè Mallorquí",
146 "Ca Rater Mallorquí": "Ca Rater Mallorquí",
147 "Cairn Terrier": "Cairn Terrier",
148 "Calupoh": "Calupoh",
149 "Campeiro Bulldog": "Campeiro Bulldog",
150 "Can de Chira": "Can de Chira",
151 "Can de Palleiro": "Can de Palleiro",
152 "Canaan Dog": "Canaan Dog",
153 "Canadian Eskimo Dog": "Canadian Eskimo Dog",
154 "Cane Corso": "Cane Corso",
155 "Cane di Oropa": "Cane di Oropa",
156 "Cane Paratore": "Cane Paratore",
157 "Cantabrian Water Dog": "Cantabrian Water Dog",
158 "Cão de Gado Transmontano": "Cão de Gado Transmontano",
159 "Cardigan Welsh Corgi": "Cardigan Welsh Corgi",
160 "Carea Leonés": "Carea Leonés",
161 "Carolina Dog": "Carolina Dog",
162 "Carpathian Shepherd Dog": "Carpathian Shepherd Dog",
163 "Castro Laboreiro Dog": "Castro Laboreiro Dog",

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164     "Catahoula Leopard Dog": "Catahoula Leopard Dog",
165     "Catalan Sheepdog": "Catalan Sheepdog",
166     "Caucasian Shepherd Dog": "Caucasian Shepherd Dog",
167     "Cavachon": "Cavachon",
168     "Cavalier King Charles Spaniel": "Cavalier King Charles
    ↪ Spaniel",
169     "Cavapoo (Cavoodle)": "Cavapoo (Cavoodle)",
170     "Central Asian Shepherd Dog": "Central Asian Shepherd Dog",
171     "Český Fousek": "Český Fousek",
172     "Český Terrier": "Český Terrier",
173     "Chesapeake Bay Retriever": "Chesapeake Bay Retriever",
174     "Chien Français Blanc et Noir": "Chien Français Blanc et
    ↪ Noir",
175     "Chien Français Blanc et Orange": "Chien Français Blanc et
    ↪ Orange",
176     "Chien Français Tricolore": "Chien Français Tricolore",
177     "Chien gris de Saint-Louis": "Chien gris de Saint-Louis",
178     "Chihuahua": "Chihuahua",
179     "Chilean Terrier": "Chilean Terrier",
180     "Chinese Crested Dog": "Chinese Crested Dog",
181     "Chinook": "Chinook",
182     "Chipin": "Chipin",
183     "Chippiparai": "Chippiparai",
184     "Chiribaya Dog": "Chiribaya Dog",
185     "Chiweenie": "Chiweenie",
186     "Chongqing": "Chongqing",
187     "Chorkie": "Chorkie",
188     "Chortai": "Chortai",
189     "Chow Chow": "Chow Chow",
190     "Chug": "Chug",
191     "Chukotka Sled Dog": "Chukotka Sled Dog",
192     "Cimarrón Uruguayo": "Cimarrón Uruguayo",
193     "Cirneco dell'Etna": "Cirneco dell'Etna",
194     "Clumber Spaniel": "Clumber Spaniel",
195     "Cockapoo": "Cockapoo",
196     "Colombian Fino Hound": "Colombian Fino Hound",
197     "Continental Bulldog": "Continental Bulldog",
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198 "Córdoba Fighting Dog": "Córdoba Fighting Dog",
199 "Corsican Dog": "Corsican Dog",
200 "Coton de Tulear": "Coton de Tulear",
201 "Cretan Hound": "Cretan Hound",
202 "Croatian Sheepdog": "Croatian Sheepdog",
203 "Cumberland Sheepdog": "Cumberland Sheepdog",
204 "Cur": "Cur",
205 "Curly-coated Retriever": "Curly-coated Retriever",
206 "Czechoslovakian Wolfdog": "Czechoslovakian Wolfdog",
207 "Dachshund": "Dachshund",
208 "Dalbo": "Dalbo",
209 "Dalmatian": "Dalmatian",
210 "Dandie Dinmont Terrier": "Dandie Dinmont Terrier",
211 "Danish Spitz": "Danish Spitz",
212 "Danish-Swedish Farmdog": "Danish-Swedish Farmdog",
213 "Denmark Feist": "Denmark Feist",
214 "Dikkulak": "Dikkulak",
215 "Dingo": "Dingo",
216 "Dobermann": "Dobermann",
217 "Dogo Argentino": "Dogo Argentino",
218 "Dogo Cubano": "Dogo Cubano",
219 "Dogo Sardesco": "Dogo Sardesco",
220 "Dogue Brasileiro": "Dogue Brasileiro",
221 "Dogue de Bordeaux": "Dogue de Bordeaux",
222 "Donggyeongi": "Donggyeongi",
223 "Dorgi": "Dorgi",
224 "Drentse Patrijshond": "Drentse Patrijshond",
225 "Drever": "Drever",
226 "Dumfriesshire Hound": "Dumfriesshire Hound",
227 "Dunker": "Dunker",
228 "Dutch Shepherd": "Dutch Shepherd",
229 "Dutch Smoushond": "Dutch Smoushond",
230 "East European Shepherd": "East European Shepherd",
231 "East Siberian Laika": "East Siberian Laika",
232 "Ecuadorian Hairless Dog": "Ecuadorian Hairless Dog",
233 "English Cocker Spaniel": "English Cocker Spaniel",
234 "English Foxhound": "English Foxhound",

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235     "English Mastiff": "English Mastiff",
236     "English Setter": "English Setter",
237     "English Shepherd": "English Shepherd",
238     "English Springer Spaniel": "English Springer Spaniel",
239     "English Toy Terrier (Black & Tan)": "English Toy Terrier
    ↪ (Black & Tan)",
240     "English Water Spaniel": "English Water Spaniel",
241     "English White Terrier": "English White Terrier",
242     "Entlebucher Mountain Dog": "Entlebucher Mountain Dog",
243     "Erbi Txakur": "Erbi Txakur",
244     "Estonian Hound": "Estonian Hound",
245     "Estrela Mountain Dog": "Estrela Mountain Dog",
246     "Eurasier": "Eurasier",
247     "Eurohound": "Eurohound",
248     "Faroese Sheepdog": "Faroese Sheepdog",
249     "Feist": "Feist",
250     "Field Spaniel": "Field Spaniel",
251     "Fila Brasileiro": "Fila Brasileiro",
252     "Finnish Hound": "Finnish Hound",
253     "Finnish Lapphund": "Finnish Lapphund",
254     "Finnish Spitz": "Finnish Spitz",
255     "Flat-coated Retriever": "Flat-coated Retriever",
256     "Florida Cracker Cur": "Florida Cracker Cur",
257     "French Bulldog": "French Bulldog",
258     "French Spaniel": "French Spaniel",
259     "Fu Quan": "Fu Quan",
260     "Fuegian": "Fuegian",
261     "Galgo Español": "Galgo Español",
262     "Gascon Saintongeais": "Gascon Saintongeais",
263     "Gaucho Sheepdog": "Gaucho Sheepdog",
264     "Georgian Shepherd": "Georgian Shepherd",
265     "Gerberian Shepsky": "Gerberian Shepsky",
266     "German Hound": "German Hound",
267     "German Longhaired Pointer": "German Longhaired Pointer",
268     "German Pinscher": "German Pinscher",
269     "German Roughhaired Pointer": "German Roughhaired Pointer",
270     "German Shepherd": "German Shepherd",
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271 "German Shorthaired Pointer": "German Shorthaired Pointer",
272 "German Spaniel": "German Spaniel",
273 "German Spitz": "German Spitz",
274 "German Wirehaired Pointer": "German Wirehaired Pointer",
275 "Giant Schnauzer": "Giant Schnauzer",
276 "Glen of Imaal Terrier": "Glen of Imaal Terrier",
277 "Goldador": "Goldador",
278 "Golden Retriever": "Golden Retriever",
279 "Goldendoodle": "Goldendoodle",
280 "Gończy Polski": "Gończy Polski",
281 "Gordon Setter": "Gordon Setter",
282 "Grand Anglo-Français Blanc et Noir": "Grand Anglo-Français
↪ Blanc et Noir",
283 "Grand Anglo-Français Blanc et Orange": "Grand Anglo-Français
↪ Blanc et Orange",
284 "Grand Anglo-Français Tricolore": "Grand Anglo-Français
↪ Tricolore",
285 "Grand Basset Griffon Vendéen": "Grand Basset Griffon
↪ Vendéen",
286 "Grand Bleu de Gascogne": "Grand Bleu de Gascogne",
287 "Grand Fauve de Bretagne": "Grand Fauve de Bretagne",
288 "Grand Griffon Vendéen": "Grand Griffon Vendéen",
289 "Great Dane": "Great Dane",
290 "Greater Swiss Mountain Dog": "Greater Swiss Mountain Dog",
291 "Greek Harehound": "Greek Harehound",
292 "Greek Shepherd": "Greek Shepherd",
293 "Greenland Dog": "Greenland Dog",
294 "Greyhound": "Greyhound",
295 "Greyster": "Greyster",
296 "Griffon Bleu de Gascogne": "Griffon Bleu de Gascogne",
297 "Griffon Bruxellois": "Griffon Bruxellois",
298 "Griffon Fauve de Bretagne": "Griffon Fauve de Bretagne",
299 "Griffon Nivernais": "Griffon Nivernais",
300 "Gull Dong": "Gull Dong",
301 "Gull Terrier": "Gull Terrier",
302 "Halden Hound": "Halden Hound",
303 "Hällefors Elkhound": "Hällefors Elkhound",


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304     "Halls Heeler": "Halls Heeler",
305     "Hamiltonstövare": "Hamiltonstövare",
306     "Hanover Hound": "Hanover Hound",
307     "Hare Indian Dog": "Hare Indian Dog",
308     "Harrier": "Harrier",
309     "Havanese": "Havanese",
310     "Hawaiian Poi Dog": "Hawaiian Poi Dog",
311     "Himalayan Sheepdog (Bhotia)": "Himalayan Sheepdog (Bhotia)",
312     "Hmong Bobtail Dog": "Hmong Bobtail Dog",
313     "Hokkaido": "Hokkaido",
314     "Hovawart": "Hovawart",
315     "Huntaway": "Huntaway",
316     "Hygen Hound": "Hygen Hound",
317     "Ibizan Hound": "Ibizan Hound",
318     "Icelandic Sheepdog": "Icelandic Sheepdog",
319     "Indian Pariah Dog": "Indian Pariah Dog",
320     "Indian Spitz": "Indian Spitz",
321     "Irish Red and White Setter": "Irish Red and White Setter",
322     "Irish Setter": "Irish Setter",
323     "Irish Terrier": "Irish Terrier",
324     "Irish Water Spaniel": "Irish Water Spaniel",
325     "Irish Wolfhound": "Irish Wolfhound",
326     "Istrian Coarse-haired Hound": "Istrian Coarse-haired Hound",
327     "Istrian Shorthaired Hound": "Istrian Shorthaired Hound",
328     "Italian Greyhound": "Italian Greyhound",
329     "Jack Russell Terrier": "Jack Russell Terrier",
330     "Jackabee": "Jackabee",
331     "Jagdterrier": "Jagdterrier",
332     "Jämthund": "Jämthund",
333     "Japanese Chin": "Japanese Chin",
334     "Japanese Spitz": "Japanese Spitz",
335     "Japanese Terrier": "Japanese Terrier",
336     "Jeju": "Jeju",
337     "Jindo": "Jindo",
338     "Jonangi": "Jonangi",
339     "Jug": "Jug",
340     "Kai Ken": "Kai Ken",
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341 "Kaikadi": "Kaikadi",
342 "Kamchatka Sled Dog": "Kamchatka Sled Dog",
343 "Kangal Shepherd Dog": "Kangal Shepherd Dog",
344 "Kangaroo Dog": "Kangaroo Dog",
345 "Kanni": "Kanni",
346 "Karakachan": "Karakachan",
347 "Karelian Bear Dog": "Karelian Bear Dog",
348 "Karelo-Finnish Laika": "Karelo-Finnish Laika",
349 "Kars": "Kars",
350 "Karst Shepherd": "Karst Shepherd",
351 "Kazakh Tazy": "Kazakh Tazy",
352 "Keeshond": "Keeshond",
353 "Kerry Beagle": "Kerry Beagle",
354 "Kerry Blue Terrier": "Kerry Blue Terrier",
355 "Khala": "Khala",
356 "King Charles Spaniel": "King Charles Spaniel",
357 "King Shepherd": "King Shepherd",
358 "King's White Hound": "King's White Hound",
359 "Kintamani": "Kintamani",
360 "Kishu": "Kishu",
361 "Kokoni": "Kokoni",
362 "Kombai": "Kombai",
363 "Komondor": "Komondor",
364 "Kooikerhondje": "Kooikerhondje",
365 "Koolie": "Koolie",
366 "Kromfohrländer": "Kromfohrländer",
367 "Kuchi": "Kuchi",
368 "Kunma": "Kunma",
369 "Kunming": "Kunming",
370 "Kurdish Mastiff": "Kurdish Mastiff",
371 "Kurī": "Kurī",
372 "Kuvasz": "Kuvasz",
373 "Labradoodle": "Labradoodle",
374 "Labrador Retriever": "Labrador Retriever",
375 "Laconian": "Laconian",
376 "Lagotto Romagnolo": "Lagotto Romagnolo",
377 "Lài": "Lài",

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378     "Laizhou Hong": "Laizhou Hong",
379     "Lakeland Terrier": "Lakeland Terrier",
380     "Lancashire Heeler": "Lancashire Heeler",
381     "Landseer": "Landseer",
382     "Langqing": "Langqing",
383     "Lapponian Herder": "Lapponian Herder",
384     "Lapponian Shepherd": "Lapponian Shepherd",
385     "Large Münsterländer": "Large Münsterländer",
386     "Leonberger": "Leonberger",
387     "Levriero Sardo": "Levriero Sardo",
388     "Lhasa Apso": "Lhasa Apso",
389     "Liangshan Dog": "Liangshan Dog",
390     "Limer": "Limer",
391     "Lithuanian Hound": "Lithuanian Hound",
392     "Lobito Herreño": "Lobito Herreño",
393     "Longdog": "Longdog",
394     "Löwchen": "Löwchen",
395     "Lucas Terrier": "Lucas Terrier",
396     "Lupo Italiano": "Lupo Italiano",
397     "Lurcher": "Lurcher",
398     "Mackenzie River Husky": "Mackenzie River Husky",
399     "Magyar Agár": "Magyar Agár",
400     "Mahratta Hound": "Mahratta Hound",
401     "Majorca Shepherd Dog": "Majorca Shepherd Dog",
402     "Mal-shi": "Mal-shi",
403     "Maltese": "Maltese",
404     "Maltipoo": "Maltipoo",
405     "Manchester Terrier": "Manchester Terrier",
406     "Maneto": "Maneto",
407     "Maremmano-Abruzzese Sheepdog": "Maremmano-Abruzzese
    ↪ Sheepdog",
408     "Markiesje": "Markiesje",
409     "Marquesan Dog": "Marquesan Dog",
410     "McNab": "McNab",
411     "Miniature American Shepherd": "Miniature American Shepherd",
412     "Miniature Bull Terrier": "Miniature Bull Terrier",
413     "Miniature Fox Terrier": "Miniature Fox Terrier",
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414         "Miniature Pinscher": "Miniature Pinscher",
415         "Miniature Schnauzer": "Miniature Schnauzer",
416         "Mixed Breed": "Mixed Breed",
417         "Mojee": "Mojee",
418         "Molossus": "Molossus",
419         "Molossus of Epirus": "Molossus of Epirus",
420         "Montenegrin Mountain Hound": "Montenegrin Mountain Hound",
421         "Morkie": "Morkie",
422         "Moscow Watchdog": "Moscow Watchdog",
423         "Moscow Water Dog": "Moscow Water Dog",
424         "Mountain Cur": "Mountain Cur",
425         "Mountain Feist": "Mountain Feist",
426         "Mudhol Hound": "Mudhol Hound",
427         "Mudi": "Mudi",
428         "Neapolitan Mastiff": "Neapolitan Mastiff",
429         "Nenets Herding Laika": "Nenets Herding Laika",
430         "New Guinea Singing Dog": "New Guinea Singing Dog",
431         "New Zealand Heading Dog": "New Zealand Heading Dog",
432         "Newfoundland": "Newfoundland",
433         "Norfolk Spaniel": "Norfolk Spaniel",
434         "Norfolk Terrier": "Norfolk Terrier",
435         "Norman Hound": "Norman Hound",
436         "Norrbottenspets": "Norrbottenspets",
437         "North Country Beagle": "North Country Beagle",
438         "Northern Inuit Dog": "Northern Inuit Dog",
439         "Norwegian Buhund": "Norwegian Buhund",
440         "Norwegian Elkhound": "Norwegian Elkhound",
441         "Norwegian Lundehund": "Norwegian Lundehund",
442         "Norwich Terrier": "Norwich Terrier",
443         "Nova Scotia Duck Tolling Retriever": "Nova Scotia Duck
444         ↪ Tolling Retriever",
445         "Nureongi": "Nureongi",
446         "Old Croatian Sighthound": "Old Croatian Sighthound",
447         "Old Danish Pointer": "Old Danish Pointer",
448         "Old English Bulldog": "Old English Bulldog",
449         "Old English Sheepdog": "Old English Sheepdog",
         "Old Spanish Pointer": "Old Spanish Pointer",
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450     "Old Welsh Grey Sheepdog": "Old Welsh Grey Sheepdog",
451     "Olde English Bulldogge": "Olde English Bulldogge",
452     "Otterhound": "Otterhound",
453     "Pachón Navarro": "Pachón Navarro",
454     "Paisley Terrier": "Paisley Terrier",
455     "Pampas Deerhound": "Pampas Deerhound",
456     "Papillon": "Papillon",
457     "Parson Russell Terrier": "Parson Russell Terrier",
458     "Pastor Garafiano": "Pastor Garafiano",
459     "Pastore della Lessinia e del Lagorai": "Pastore della
    ↪ Lessinia e del Lagorai",
460     "Patagonian Sheepdog": "Patagonian Sheepdog",
461     "Patterdale Terrier": "Patterdale Terrier",
462     "Pekepo": "Pekepo",
463     "Pekingese": "Pekingese",
464     "Pembroke Welsh Corgi": "Pembroke Welsh Corgi",
465     "Perdigueiro Galego": "Perdigueiro Galego",
466     "Perro Majorero": "Perro Majorero",
467     "Perro Montero Valdueza": "Perro Montero Valdueza",
468     "Peruvian Hairless Dog": "Peruvian Hairless Dog",
469     "Petit Basset Griffon Vendéen": "Petit Basset Griffon
    ↪ Vendéen",
470     "Petit Bleu de Gascogne": "Petit Bleu de Gascogne",
471     "Phalène": "Phalène",
472     "Pharaoh Hound": "Pharaoh Hound",
473     "Philippine Forest Dog": "Philippine Forest Dog",
474     "Phu Quoc Ridgeback": "Phu Quoc Ridgeback",
475     "Picardy Spaniel": "Picardy Spaniel",
476     "Plott Hound": "Plott Hound",
477     "Plummer Terrier": "Plummer Terrier",
478     "Podenco Andaluz": "Podenco Andaluz",
479     "Podenco Canario": "Podenco Canario",
480     "Podenco Valenciano": "Podenco Valenciano",
481     "Pointer": "Pointer",
482     "Poitevin": "Poitevin",
483     "Polish Greyhound": "Polish Greyhound",
484     "Polish Hound": "Polish Hound",
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485 "Polish Lowland Sheepdog": "Polish Lowland Sheepdog",
486 "Polynesian Dog": "Polynesian Dog",
487 "Pomchi": "Pomchi",
488 "Pomeranian": "Pomeranian",
489 "Pont-Audemer Spaniel": "Pont-Audemer Spaniel",
490 "Poochon": "Poochon",
491 "Poodle": "Poodle",
492 "Porcelaine": "Porcelaine",
493 "Portuguese Podengo": "Portuguese Podengo",
494 "Portuguese Pointer": "Portuguese Pointer",
495 "Portuguese Sheepdog": "Portuguese Sheepdog",
496 "Portuguese Water Dog": "Portuguese Water Dog",
497 "Posavac Hound": "Posavac Hound",
498 "Pražský Krysařík": "Pražský Krysařík",
499 "Presa Canario": "Presa Canario",
500 "Pudelpointer": "Pudelpointer",
501 "Pug": "Pug",
502 "Pugese": "Pugese",
503 "Puggle": "Puggle",
504 "Puli": "Puli",
505 "Pumi": "Pumi",
506 "Pungsan": "Pungsan",
507 "Pyrenean Mastiff": "Pyrenean Mastiff",
508 "Pyrenean Mountain Dog": "Pyrenean Mountain Dog",
509 "Pyrenean Sheepdog": "Pyrenean Sheepdog",
510 "Rafeiro do Alentejo": "Rafeiro do Alentejo",
511 "Rajapalayam": "Rajapalayam",
512 "Rampur Greyhound": "Rampur Greyhound",
513 "Rastreador Brasileiro": "Rastreador Brasileiro",
514 "Rat Terrier": "Rat Terrier",
515 "Ratonero Murciano": "Ratonero Murciano",
516 "Redbone Coonhound": "Redbone Coonhound",
517 "Rhodesian Ridgeback": "Rhodesian Ridgeback",
518 "Rize Koyun": "Rize Koyun",
519 "Romanian Mioritic Shepherd Dog": "Romanian Mioritic Shepherd
↪ Dog",
520 "Romanian Raven Shepherd Dog": "Romanian Raven Shepherd Dog",

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521     "Rottweiler": "Rottweiler",
522     "Rough Collie": "Rough Collie",
523     "Russian Spaniel": "Russian Spaniel",
524     "Russkiy Toy": "Russkiy Toy",
525     "Russo-European Laika": "Russo-European Laika",
526     "Ryukyu": "Ryukyu",
527     "Saarloos Wolfdog": "Saarloos Wolfdog",
528     "Sabueso Español": "Sabueso Español",
529     "Saint Miguel Cattle Dog": "Saint Miguel Cattle Dog",
530     "Saint-Usuge Spaniel": "Saint-Usuge Spaniel",
531     "Sakhalin Husky": "Sakhalin Husky",
532     "Salish Wool Dog": "Salish Wool Dog",
533     "Saluki": "Saluki",
534     "Samoyed": "Samoyed",
535     "Sapsali": "Sapsali",
536     "Sarabi": "Sarabi",
537     "Sarail Hound": "Sarail Hound",
538     "Sardinian Shepherd Dog": "Sardinian Shepherd Dog",
539     "Šarplaninac": "Šarplaninac",
540     "Schapendoes": "Schapendoes",
541     "Schillerstövare": "Schillerstövare",
542     "Schipperke": "Schipperke",
543     "Schnoodle": "Schnoodle",
544     "Schweizer Laufhund": "Schweizer Laufhund",
545     "Schweizerischer Niederlaufhund": "Schweizerischer
    ↪  Niederlaufhund",
546     "Scottish Deerhound": "Scottish Deerhound",
547     "Scottish Terrier": "Scottish Terrier",
548     "Sealyham Terrier": "Sealyham Terrier",
549     "Segugio dell'Appennino": "Segugio dell'Appennino",
550     "Segugio Italiano": "Segugio Italiano",
551     "Segugio Maremmano": "Segugio Maremmano",
552     "Serbian Hound": "Serbian Hound",
553     "Serbian Tricolour Hound": "Serbian Tricolour Hound",
554     "Serrano Bulldog": "Serrano Bulldog",
555     "Shar Pei": "Shar Pei",
556     "Sheepadoodle": "Sheepadoodle",
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557 "Shetland Sheepdog": "Shetland Sheepdog",
558 "Shiba Inu": "Shiba Inu",
559 "Shih Tzu": "Shih Tzu",
560 "Shih-poo": "Shih-poo",
561 "Shikoku": "Shikoku",
562 "Shiloh Shepherd": "Shiloh Shepherd",
563 "Siberian Husky": "Siberian Husky",
564 "Silken Windhound": "Silken Windhound",
565 "Sinhala Hound": "Sinhala Hound",
566 "Skye Terrier": "Skye Terrier",
567 "Sloughi": "Sloughi",
568 "Slovak Rough-haired Pointer": "Slovak Rough-haired Pointer",
569 "Slovenský Čuvač": "Slovenský Čuvač",
570 "Slovenský Kopov": "Slovenský Kopov",
571 "Smaland Hound": "Smaland Hound",
572 "Small Međimurje Dog": "Small Međimurje Dog",
573 "Small Münsterländer": "Small Münsterländer",
574 "Smithfield": "Smithfield",
575 "Smooth Collie": "Smooth Collie",
576 "Smooth Fox Terrier": "Smooth Fox Terrier",
577 "Soft-coated Wheaten Terrier": "Soft-coated Wheaten Terrier",
578 "South Russian Ovcharka": "South Russian Ovcharka",
579 "Southern Hound": "Southern Hound",
580 "Spanish Mastiff": "Spanish Mastiff",
581 "Spanish Water Dog": "Spanish Water Dog",
582 "Spino degli Iblei": "Spino degli Iblei",
583 "Spinone Italiano": "Spinone Italiano",
584 "Sporting Lucas Terrier": "Sporting Lucas Terrier",
585 "Springador": "Springador",
586 "St. Bernard": "St. Bernard",
587 "St. Hubert Jura Hound": "St. Hubert Jura Hound",
588 "St. John's Water Dog": "St. John's Water Dog",
589 "Stabyhoun": "Stabyhoun",
590 "Staffordshire Bull Terrier": "Staffordshire Bull Terrier",
591 "Staghound": "Staghound",
592 "Standard Schnauzer": "Standard Schnauzer",
593 "Stephens Stock": "Stephens Stock",


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594     "Styrian Coarse-haired Hound": "Styrian Coarse-haired Hound",
595     "Sussex Spaniel": "Sussex Spaniel",
596     "Swedish Lapphund": "Swedish Lapphund",
597     "Swedish Vallhund": "Swedish Vallhund",
598     "Tahitian Dog": "Tahitian Dog",
599     "Tahltan Bear Dog": "Tahltan Bear Dog",
600     "Taigan": "Taigan",
601     "Taiwan Dog": "Taiwan Dog",
602     "Talbot": "Talbot",
603     "Tamaskan Dog": "Tamaskan Dog",
604     "Tang Dog": "Tang Dog",
605     "Tarsus Çatalburun": "Tarsus Çatalburun",
606     "Tatra Shepherd Dog": "Tatra Shepherd Dog",
607     "Techichi": "Techichi",
608     "Teddy Roosevelt Terrier": "Teddy Roosevelt Terrier",
609     "Telomian": "Telomian",
610     "Tenterfield Terrier": "Tenterfield Terrier",
611     "Terceira Mastiff": "Terceira Mastiff",
612     "Tesem": "Tesem",
613     "Texas Heeler": "Texas Heeler",
614     "Thai Bangkaew Dog": "Thai Bangkaew Dog",
615     "Thai Ridgeback": "Thai Ridgeback",
616     "Tibetan Kyi Apso": "Tibetan Kyi Apso",
617     "Tibetan Mastiff": "Tibetan Mastiff",
618     "Tibetan Spaniel": "Tibetan Spaniel",
619     "Tibetan Terrier": "Tibetan Terrier",
620     "Tonya Finosu (Trabzon Finosu)": "Tonya Finosu (Trabzon
    ↪  Finosu)",
621     "Tornjak": "Tornjak",
622     "Tosa": "Tosa",
623     "Toy Bulldog": "Toy Bulldog",
624     "Toy Fox Terrier": "Toy Fox Terrier",
625     "Toy Manchester Terrier": "Toy Manchester Terrier",
626     "Toy Trawler Spaniel": "Toy Trawler Spaniel",
627     "Transylvanian Hound": "Transylvanian Hound",
628     "Treeing Cur": "Treeing Cur",
629     "Treeing Feist": "Treeing Feist",
```

630 "Treeing Tennessee Brindle": "Treeing Tennessee Brindle",
631 "Treeing Walker Coonhound": "Treeing Walker Coonhound",
632 "Trigg Hound": "Trigg Hound",
633 "Turnspit": "Turnspit",
634 "Tweed Water Spaniel": "Tweed Water Spaniel",
635 "Tyrolean Hound": "Tyrolean Hound",
636 "Valencian Terrier": "Valencian Terrier",
637 "Vikhan Sheepdog": "Vikhan Sheepdog",
638 "Villano de Las Encartaciones": "Villano de Las
↪ Encartaciones",
639 "Villanuco de Las Encartaciones": "Villanuco de Las
↪ Encartaciones",
640 "Vizsla": "Vizsla",
641 "Volkosob": "Volkosob",
642 "Volpino Italiano": "Volpino Italiano",
643 "Weimaraner": "Weimaraner",
644 "Welsh Hillman": "Welsh Hillman",
645 "Welsh Hound": "Welsh Hound",
646 "Welsh Sheepdog": "Welsh Sheepdog",
647 "Welsh Springer Spaniel": "Welsh Springer Spaniel",
648 "Welsh Terrier": "Welsh Terrier",
649 "West Country Harrier": "West Country Harrier",
650 "West Highland White Terrier": "West Highland White Terrier",
651 "West Siberian Laika": "West Siberian Laika",
652 "Westiepoo": "Westiepoo",
653 "Westphalian Dachsbracke": "Westphalian Dachsbracke",
654 "Wetterhoun": "Wetterhoun",
655 "Whippet": "Whippet",
656 "White Shepherd": "White Shepherd",
657 "White Swiss Shepherd Dog": "White Swiss Shepherd Dog",
658 "Whoodle": "Whoodle",
659 "Wire Fox Terrier": "Wire Fox Terrier",
660 "Wirehaired Pointing Griffon": "Wirehaired Pointing Griffon",
661 "Wirehaired Vizsla": "Wirehaired Vizsla",
662 "Xiasi Dog": "Xiasi Dog",
663 "Xigou": "Xigou",
664 "Xoloitzcuintle": "Xoloitzcuintle",

```
665         "Yakutian Laika": "Yakutian Laika",
666         "Yorkiepool": "Yorkiepool",
667         "Yorkshire Terrier": "Yorkshire Terrier",
668         "Zerdava": "Zerdava",
669         "Zuchon": "Zuchon"
670     }
671 },
672 "sex": {
673     "LongName": "Sex",
674     "Description": "Biological sex of the participant",
675     "Format": "string",
676     "Levels": {
677         "male": "Male",
678         "female": "Female"
679     }
680 },
681 "neutered": {
682     "LongName": "Neutered",
683     "Description": "Whether the participant has been neutered or
684     ↪ spayed",
685     "Format": "string",
686     "Levels": {
687         "true": "Neutered or spayed",
688         "false": "Not neutered or spayed"
689     }
690 },
691 "age_months": {
692     "LongName": "Age (months)",
693     "Description": "Age of the participant at the time of the study
694     ↪ expressed in months",
695     "Format": "integer",
696     "Units": "months",
697     "Minimum": 0
698 },
699 "weight_kg": {
700     "LongName": "Weight (kg)",
```

```

699         "Description": "Body weight of the participant at the time of the
        ↪ study expressed in kg",
700         "Format": "number",
701         "Units": "kilograms",
702         "Minimum": 0.1
703     },
704     "daily_time_with_owner_hours": {
705         "LongName": "Daily time with owner (hours)",
706         "Description": "Amount of time in hours, that the participant dog
        ↪ spends on average with its owner per day",
707         "Format": "integer",
708         "Units": "hours",
709         "Minimum": 0,
710         "Maximum": 24
711     },
712     "provenance": {
713         "LongName": "Provenance",
714         "Description": "Origin of the participant prior to inclusion in
        ↪ the study",
715         "Format": "string",
716         "Levels": {
717             "pet store": "Obtained from a pet store",
718             "private breeder": "Obtained from a private breeder",
719             "animal shelter": "Obtained from an animal shelter",
720             "animal rescue organization": "Obtained from an animal rescue
        ↪ organization",
721             "self bred": "Bred by the current owner",
722             "other owner": "Obtained from other owner",
723             "found as a stray": "Found as a stray",
724             "other": "Other provenance"
725         }
726     }
727 }

```

B.5 scans.json

```
1 {
2   "filename": {
3     "LongName": "Filename",
4     "Description": "Relative paths to the associated file. Paths are
      ↳ relative to the session directory. Uses '/' as separator
      ↳ independently of the operative system"
5   },
6   "acq_time": {
7     "LongName": "Acquisition time",
8     "Description": "Acquisition time of the first data point of the
      ↳ file"
9   }
10 }
```

B.6 sessions.json

```
1 {
2   "session_id": {
3     "LongName": "Session identifier",
4     "Description": "Session ID"
5   },
6   "acq_time": {
7     "LongName": "Acquisition time",
8     "Description": "Acquisition time of the first data point of the
      ↳ session"
9   }
10 }
```

B.7 sub-00_sessions.tsv

1	session_id	acq_time
2	ses-pres01	2026-03-03T09:31:50.985053

B.8 sub-00_ses-pres01__scans.tsv

```

1 filename          acq_time
2 video/sub-00_ses-pres01_task-calib_run-1_recording-cam1_video.mp4 |
  ↪          2026-03-03T09:31:50.985053
3 video/sub-00_ses-pres01_task-calib_run-1_recording-cam2_video.mp4 |
  ↪          2026-03-03T09:31:50.986021
4 audio/sub-00_ses-pres01_task-calib_run-1_recording-micdog_audio.wav |
  ↪          2026-03-03T09:31:50.988822

```

B.9 sub-01_ses-scens01__task-treat__run-1__recording-ecg_physio.json

```

1 {
2   "SamplingFrequency": 100.0,
3   "StartTime": 12.5,
4   "Columns": [
5     "timestamp",
6     "ecg_1",
7     "ecg_2"
8   ],
9   "PhysioType": "generic",
10  "timestamp": {
11    "LongName": "Timestamp",
12    "Description": "Timestamp of the ECG recording, expressed in
  ↪ millisecond, starting from the moment the first data point of the
  ↪ recording was acquired. The first timestamp is 0.0",
13    "Format": "number",
14    "Units": "seconds since the beginning of the recording",
15    "Minimum": 0.0
16  },
17  "ecg_1": {
18    "LongName": "ECG Recording 1",
19    "Description": "continuous ECG recording 1",
20    "Format": "number",

```

```

21     "Units": "mV"
22   },
23   "ecg_2": {
24     "LongName": "ECG Recording 2",
25     "Description": "continuous ECG recording 2",
26     "Format": "number",
27     "Units": "mV"
28   }
29 }

```

B.10 sub-01_ses-scens01_task-treat_tracksys-mpu_run-1_motion.json

```

1  {
2    "SamplingFrequency": 300.0,
3    "SamplingFrequencyUnit": "Hz",
4    "TaskName": "treat",
5    "InstitutionName": "TUAT",
6    "InstitutionAddress": "Japan, Tokyo, Koganei",
7    "InstitutionalDepartmentName": "Department of Computer Engineering"
8  }

```

B.11 sub-01_ses-scens01_task-treat_tracksys-mpu_run-1_channels.tsv

1	name	component	type	tracked_point	units
2	timestamp	n/a	LATENCY	n/a	s
3	mpu1_ax	x	ACCEL	chest	m/s ²
4	mpu1_ay	y	ACCEL	chest	m/s ²
5	mpu1_az	z	ACCEL	chest	m/s ²
6	mpu1_gx	x	GYRO	chest	rad/s
7	mpu1_gy	y	GYRO	chest	rad/s
8	mpu1_gz	z	GYRO	chest	rad/s
9	mpu2_ax	x	ACCEL	chest	m/s ²

10	mpu2_ay	y	ACCEL	chest	m/s ²
11	mpu2_az	z	ACCEL	chest	m/s ²
12	mpu2_gx	x	GYRO	chest	rad/s
13	mpu2_gy	y	GYRO	chest	rad/s
14	mpu2_gz	z	GYRO	chest	rad/s
15	mpu3_ax	x	ACCEL	chest	m/s ²
16	mpu3_ay	y	ACCEL	chest	m/s ²
17	mpu3_az	z	ACCEL	chest	m/s ²
18	mpu3_gx	x	GYRO	chest	rad/s
19	mpu3_gy	y	GYRO	chest	rad/s
20	mpu3_gz	z	GYRO	chest	rad/s
21	mpu4_ax	x	ACCEL	chest	m/s ²
22	mpu4_ay	y	ACCEL	chest	m/s ²
23	mpu4_az	z	ACCEL	chest	m/s ²
24	mpu4_gx	x	GYRO	chest	rad/s
25	mpu4_gy	y	GYRO	chest	rad/s
26	mpu4_gz	z	GYRO	chest	rad/s

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